THE INCOMPLETENESS OF OBSERVATION

The Equivalence of Quantum Mechanics and Embedded Observation

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ABSTRACT

An observer embedded in a deterministic universe cannot access the complete state. We prove that any such observer — coupled to a slow, high-capacity hidden sector on a finite-dimensional configuration space — must describe the visible sector using P-indivisible stochastic dynamics. The first half of the claim is established here in full: marginalizing a deterministic bijection over an inaccessible hidden sector yields a generically non-Markovian (P-indivisible) visible process, and this is the framework's own contribution. The identification of that process with unitary quantum mechanics is supplied by the Barandes stochastic-quantum correspondence [11, 12], an external result that we use as a load-bearing dependency rather than re-derive: it furnishes a (basis-dependent, non-unique) representation of any P-indivisible process on a finite configuration space as a unitary system whose transition probabilities are exactly Born-rule probabilities. The correspondence delivers the kinematic scaffolding and transition statistics of quantum mechanics — complex Hilbert space, unitary evolution, the Born rule — while non-commuting observables and spin are introduced as additional structure rather than derived from the bare stochastic data, with the composite-system tensor-product structure following instead, for spatially separated subsystems, from the substratum's coupling-graph locality (§3.1, scope remark; §3.4). The converse direction (any quantum system, realized as a deterministic dilation, requires non-trivial coupling, slow-bath memory, and sufficient hidden-sector capacity) holds with the C2 necessity step conditional on the eigenstate thermalization hypothesis in the hidden sector (§3.4). The Schrödinger equation, Born rule, and Bell violations emerge as structural consequences within this representation, requiring no independent quantum postulates beyond the correspondence itself. The framework's posture is therefore one of *identification* — quantum mechanics as the description of a particular kind of incomplete observation — rather than derivation of quantum mechanics from a parameter-free first principle; the distinction is the proper locus of skeptical scrutiny.

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1. INTRODUCTION

1.1 The Problem of Embedded Observation

An observer embedded in a deterministic universe cannot access the complete state: degrees of freedom beyond causal reach are permanently hidden, and the observer's description is obtained by marginalizing over the hidden sector. Whether a description obtained this way can be considered complete is the question Einstein, Podolsky, and Rosen posed of quantum mechanics [46]; it returns here with the opposite valence — incompleteness is derived as a structural property of embedded observation, and quantum mechanics is identified as its form. The question is what this reduction imposes on the form of the observer's physical laws.

Prior work (QBism, relational QM, 't Hooft's cellular automaton [1]) takes observer-dependence as an interpretive starting point or derives it from specific microphysical models. This framework differs by identifying *necessary and sufficient conditions* under which any embedded observer in any deterministic system necessarily sees quantum mechanics — and proving these are the *only* conditions under which QM arises from a deterministic embedding.

1.2 The Starting Point

The framework begins from a starting point usually stated in two words: *observation occurs*. An observer records distinguishable outcomes of interactions with a system not wholly under the observer's control. This is the cogito of Descartes made precise — not as philosophy, but as a mathematical constraint. As §1.2 makes explicit, that starting point is not a single indubitable fact but resolves into two axioms, one indubitable and one a substantive posit; the two-word phrase is the informal name for the pair.

The axiom has structural content beyond contingent occurrence. Substrata that do not admit embedded observers exist as mathematical objects with dynamics; they do not produce physics in the present framework's sense. The axiom thus commits to our universe being in the observer-admitting subset of substrata — substrata whose bijection structure satisfies the conditions C1–C3 below for some partition. This is a structural commitment about the framework's domain, not a contingent fact about whether someone happens to be looking.

The phrase "observation occurs" can be stated with more precision than a single undifferentiated axiom. Distilled, it is *two axioms.* **Axiom 1** is the evidential floor: tokened differentiation occurs — concrete differentiated content is *registered*, a this-not-that tokened from a locus against a remainder; equivalently, *embedded representation occurs.* It is indubitable, since to doubt it is itself to register a distinction from a locus — the cogito made precise, which is irreducibly perspectival: a doubting *I*, not a free-floating fact. Read this way, the primitive already contains three things that therefore need no separate derivation: a differentiation (the this-not-that), a locus that registers it (the observer), and the remainder it is registered against. The proper-part decomposition $V \subsetneq S$ — the observer embedded in a whole it does not exhaust — is accordingly constitutive of the primitive rather than a downstream posit: a registering that left no remainder would be the whole representing itself, not an observation. From Axiom 1, by analysis of the words "occurs" and "registered," the relational structure, the existence of a dynamics, determinism, finiteness per moment, and bijectivity follow as lemmas — they are not independent assumptions. One thing does not follow: that the differentiation *recurs* — that there is more than one moment for the dynamics to act across. This is **Axiom 2**: differentiation recurs, the temporal domain being the minimal such recurrence. Axiom 2 presupposes Axiom 1 (there is no recurrence of differentiation without differentiation) but is not derivable from it: a structure containing a single tokened differentiation and nothing further satisfies Axiom 1 while containing no recurrence, so no analysis of Axiom 1 alone can yield Axiom 2 — non-derivability by countermodel. The framework's foundation is therefore most honestly described not as a single axiom but as a *two-axiom observation base*: one indubitable claim, and one substantive posit that the first does not yield. The two invoke a single principle — tokened differentiation — and are thematically one; they are numerically two. The Definition and Lemmas below unpack this structure.

The perspectival reading of the primitive is a commitment, and it is worth stating what the alternative would cost. One could instead read the evidential floor *thinly* — as bare differentiated content with no locus occupying a side — in which case the embeddedness $V \subsetneq S$ does not follow and would have to be added as a third, independent axiom. The framework adopts the perspectival reading: the floor the cogito secures is already a registering, not a free-standing fact, so embeddedness belongs to the primitive rather than being posited beside it. This choice has a payoff beyond economy of axioms. On the perspectival reading the foundation's incompleteness is an *internal* consequence of the primitive, not an external import: an embedded representation cannot close over the whole that contains it, since a complete self-inclusive representation would have to represent its own representing without remainder, collapsing the locus into the whole. This is the same form of diagonal that limits formal systems (Gödel [49]) and embedded inference devices (Wolpert [50]); those results corroborate the internal statement rather than supply it. The choice between the perspectival primitive and the thin-plus-third-axiom formulation is presentational — it bears only on how the foundation's commitments are counted and changes no result downstream of the structure (S, φ, V) — but the perspectival statement is the more faithful to the indubitable floor the cogito secures, which is why it is adopted here.

We ask: what is the minimal mathematical structure consistent with this fact?

Definition. An *observation* is a triple (S, φ, V): a total system S, a dynamics $\varphi : S \to S$, and an observer $V \subsetneq S$ — a proper subsystem with finitely many distinguishable internal states, coupled to the complement $H = S \setminus V$ through φ.

This definition formalizes three features implicit in the concept of observation: there is a whole (S), the observer is not the whole ($V \subsetneq S$), and the observer registers changes (coupling through φ). From this definition, three structural properties follow. The dynamics φ itself is nothing exotic, and its form need not be posited: it follows from a single innocuous premise, that evolution *composes* — advancing the state by t and then by s equals advancing it by $t + s$. This composition (semigroup) law is Cauchy's functional equation, and it forces the evolution to be a one-parameter family generated by a fixed rule [45]; in the continuous case this is the *phase flow* foundational to classical mechanics [44], and in the discrete case relevant here it collapses to iteration of a single transition map, $x_t = \varphi^t(x_0)$. Thus the state-transition structure is not an assumption particular to this framework but a consequence of consistency of time evolution — the same structure standard physics rests on. What distinguishes the present setting is not the existence of such a map but that it acts on a *finite* S (Lemma 1); finiteness, not the dynamics, is the load-bearing commitment, and it is what renders energy and its conservation well defined downstream.

Lemma 1 (Finiteness). *The observer has finitely many distinguishable internal states, so the visible configuration space $\mathcal{C}_V$ is finite, with a discreteness scale ϵ providing a finite minimal cell volume.* Any observer bounded by a finite-area surface can couple to only finitely many modes across that boundary; independent support comes from holographic entropy bounds [2].

Lemma 2 (Causal partition). *The observer is a proper subsystem: $V \subsetneq S$. The complement $H = S \setminus V$ is the hidden sector.* This partition is not a modeling choice but the definition of embedded observation: an observer that could access all of S would have nothing external to observe. The product decomposition

$$\Gamma = \Gamma_V \times \Gamma_H, \qquad H_{\text{tot}} = H_V + H_H + H_{\text{int}}$$

follows, with cross-partition correlations entering only through H_{int}. The product decomposition is idealized; §4.5 addresses approximation quality.

Lemma 3 (Determinism and unique measure). *φ is a bijection: distinct states have distinct successors (determinism) and distinct predecessors (reversibility). The counting measure is the unique φ-invariant measure on the accessible orbit, and the unique maximal-entropy invariant measure globally.* Determinism (injectivity) holds by a two-pronged argument. For states differing in registered content, injectivity is the requirement that the dynamics preserve the observer's records — merging them would erase a distinction that has occurred. For states differing only in the hidden sector, a merge is either statistically observable or removable: a merge whose successor configurations differ visibly breaks the double stochasticity of the marginalized process at some step order (the marginal of a bijection under the uniform prior is doubly stochastic by Birkhoff–von Neumann, §3.2, and unitary emergent statistics are unistochastic), so it is excluded by the observed unitarity of quantum dynamics; a merge that leaves no statistical trace is removable, since any map on a finite set restricts to a bijection on its *eventual image*, where any recurrent registered history lives — the non-injective description is then equivalent, for all registered history, to a bijective one. Either horn delivers a bijective substratum for the dynamics governing registered history; the two prongs use different inputs — the first empirical (the observed unitarity of quantum dynamics), the second structural (finiteness and recurrence). On a finite set, injectivity implies surjectivity, so φ is a bijection. As for the measure: on the orbit of the actual microstate, the uniform measure is the unique φ-invariant measure; the selection among orbits is fixed by maximal entropy compatible with the observer's records — and the atypical short orbits of the idealized linear dynamics (fixed points, low-period cycles) are precisely observer-lacking substrata — an orbit of period p makes at most p states dynamically accessible, so for small p no partition of it can satisfy the high-capacity condition C3 — admitting no C1–C3 partition, hence outside the scope of every theorem below. The invariant counting measure, in the continuum limit, becomes the Liouville measure on Γ:

$$\frac{\partial \rho}{\partial t} = \{H_{\text{tot}}, \rho\} \equiv \mathcal{L}\rho$$

Remark (reversibility as the conservative assumption). The bijectivity of φ — equivalently, that the substratum conserves information — is not an exotic commitment peculiar to this framework but the orthodox assumption of mainstream physics, imported here rather than invented. Unitary evolution in quantum mechanics *is* information conservation: a pure state evolves from a unique pure state, so the dynamics is reversible and the past is in principle recoverable from the present [42]. The classical counterpart is Liouville's theorem, the conservation of phase-space volume recovered above. That the physics community treats information conservation as effectively non-negotiable is shown by the black-hole information paradox, in which the *apparent* violation of unitarity by Hawking evaporation is regarded as a paradox to be resolved rather than a discovery to be accepted [43, 42]; the no-hiding theorem [42] makes the underlying conservation statement rigorous. The framework therefore assumes no more than standard physics already does — and it is worth noting that the literature independently records the same information↔energy link that the companion Explainer draws via Noether's theorem: a violation of unitarity would also violate energy conservation. What is non-standard here is not the assumption of reversibility but the claim to *recover* it, below, from the two-pronged argument of Lemma 3; a reader unwilling to grant that recovery may simply take reversibility as the standard postulate it is elsewhere, with no cost to the emergence results.

These properties contain no quantum postulates. The claim is that quantum mechanics *emerges* under the conditions below.

1.3 Conditions on the Hidden Sector

(C1) Non-zero coupling. $H_{\text{int}} \neq 0$. The coupling is bidirectional. (Status: doubly anchored. Necessity — observed Bell violations force C1, by the characterization theorem of §3.4. Realization — at the cosmological horizon the ADM boundary Hamiltonian and the constraint structure supply a coupling *stronger* than $H_{\text{int}} \neq 0$ [GR §2.2].)

(C2) Slow-bath timescale separation. The visible-sector (system) timescale τ_S is much shorter than the hidden-sector (bath) timescale τ_B: $\tau_S \ll \tau_B$ — the *inverse* of the Markovian regime. The hidden sector evolves on timescales far exceeding those accessible to the observer. (Status: necessity — observed QM forces C2 by §3.4, conditional on ETH for the hidden sector; ETH's own status, a well-supported conjecture rather than a theorem, is stated at the remark in §3.4. Realization — $\tau_S/\tau_B \sim 10^{-32}$ at the horizon [GR §2.2].)

(C3) Sufficient capacity. The number of hidden-sector degrees of freedom N_H exceeds the number of visible-sector degrees of freedom N_V by enough that visible-sector interactions do not appreciably perturb the hidden sector's state on timescales $\ll \tau_B$. (Status: necessity — observed information backflow forces the per-process capacity bound $m \geq 2^{I^*}$ of §3.4. Realization — $A/\epsilon^2 \sim 10^{122}$ hidden modes against $\sim 10^{80}$ baryons; no laboratory process saturates the hidden sector [GR §2.2].)

Theorem statement. Under Lemmas 1–3 and (C1)–(C3), the embedded observer's reduced description is mathematically equivalent to unitarily evolving quantum mechanics (defined precisely in §3.4). The conditions are not merely sufficient but *necessary* (§3.4), establishing full equivalence.

1.4 Partition-Relativity

Lemma. *The emergent description is uniquely determined by the partition. Any parameters of the emergent theory depend only on the geometric and thermodynamic properties of the partition boundary.*

Proof. By Lemma 3, the Hamiltonian flow ϕ_t is unique. By Lemma 2, the partition is fixed. By Lemma 3, the observer uses the Liouville measure μ — the unique absolutely continuous invariant measure on the full phase space. (An embedded observer with incomplete knowledge of the total energy averages over energies with a smooth prior, recovering Liouville on a phase-space band — the continuum form of Lemma 3's maximal-entropy selection; singular invariant measures concentrated on atypical short orbits correspond to observer-lacking substrata and are outside scope by the same lemma.) Letting π_V denote projection onto the visible sector, the marginalized transition probabilities

$$T_{ij}(t_2, t_1) = \int_{\Gamma_H} \delta_{x_j}[\pi_V(\phi_{t_2-t_1}(x_i, h))]\, d\mu(h)$$

are therefore uniquely determined by three inputs: dynamics (ϕ_t), partition (Γ_V, Γ_H), and measure (μ). Since ϕ_t and μ are fixed by the definition and lemmas, the stochastic process — and hence any emergent quantum description (§3.1) — depends only on the partition. □

2. EMERGENT STOCHASTICITY AND P-INDIVISIBILITY

2.1 Emergent Stochasticity

The total system evolves deterministically, but the visible sector alone is stochastic. The observer knows x but not h; different hidden states h_k compatible with the same x send it to different futures x'_k. Transition probabilities are obtained by marginalizing over the Liouville measure (§1.4). This makes the framework a hidden variable theory; consistency with Bell's theorem is addressed in §3.3.

2.2 The Slow-Bath Regime

The Markovian limit requires $\tau_B \ll \tau_S$ [3]. Condition (C2) inverts this: $\tau_S \ll \tau_B$. A slow bath must be distinguished from a static field. By (C1), coupling is continuously active: each visible-sector transition imprints on the hidden sector. Because the hidden sector is slow (not static), imprints persist without thermal overwriting. On subsequent transitions, coupling reads back stored correlations, producing history-dependent transition probabilities — the non-Markovian regime [3].

2.3 P-Indivisibility

By Lemma 1, the visible and hidden sectors have finite configuration spaces $\mathcal{C}_V$ and $\mathcal{C}_H$ (the discrete counterparts of Γ_V and Γ_H). On these finite sets:

Theorem. *Let $\mathcal{C}_V$ and $\mathcal{C}_H$ be finite sets with $|\mathcal{C}_V| \geq 2$, and let $\varphi : \mathcal{C}_V \times \mathcal{C}_H \to \mathcal{C}_V \times \mathcal{C}_H$ be a bijection. Define:*

$$T_{ij} = \frac{|\{h \in \mathcal{C}_H : \pi_V(\varphi(x_i, h)) = x_j\}|}{|\mathcal{C}_H|}$$

and the k-step matrix $T^{(k)}_{ij}$ by applying φ^k. If T is not a permutation matrix, then the process is P-indivisible.

Proof. Uses total variation distance $d(p, q) = \frac{1}{2}\sum_k |p_k - q_k|$. P-divisibility $\iff$ $d(p(t), q(t))$ non-increasing for all initial p, q [4, 5].

Step 1 (Recurrence). φ bijective on a finite set $\Rightarrow \exists N : \varphi^N = \mathrm{id}$. Thus $T^{(N)} = I$ and $d(\delta_i T^{(N)}, \delta_j T^{(N)}) = 1$ for $i \neq j$.

Step 2 (Strict contraction). T not a permutation $\Rightarrow \exists i, j, l$: $T_{il} > 0$ and $T_{jl} > 0$. Then:

$$d(\delta_i T, \delta_j T) = \frac{1}{2} \sum_k |T_{ik} - T_{jk}| < 1$$

Step 3. $d(\delta_i T^{(1)}, \delta_j T^{(1)}) < 1 = d(\delta_i T^{(N)}, \delta_j T^{(N)})$: non-monotonic, hence P-indivisible. □

The theorem requires only bijective dynamics (Lemma 3) and non-trivial coupling (C1). Lemma 1 guarantees the finite configuration space.

Genericity. The theorem's precondition — that T is not a permutation matrix — is equivalent to (C1) at the bijection level and is generic among bijections on $\mathcal{C}_V \times \mathcal{C}_H$ in a quantifiable sense. A bijection φ on $\mathcal{C}_V \times \mathcal{C}_H$ produces T equal to a permutation matrix if and only if φ has the form $\varphi(x_i, h) = (\varphi_V(x_i), \sigma_{x_i}(h))$ — the V-dynamics is independent of h. Counting these: $|\mathcal{C}_V|!$ choices for φ_V times $(|\mathcal{C}_H|!)^{|\mathcal{C}_V|}$ choices for the family $\{\sigma_{x_i}\}$, giving a total of $|\mathcal{C}_V|!\,(|\mathcal{C}_H|!)^{|\mathcal{C}_V|}$ bijections with T a permutation, out of $(|\mathcal{C}_V|\,|\mathcal{C}_H|)!$ total. The fraction failing the precondition is therefore

$$P_{\text{perm}}(n_V, n_H) = \frac{n_V!\,(n_H!)^{n_V}}{(n_V n_H)!}, \quad n_V := |\mathcal{C}_V|,\ n_H := |\mathcal{C}_H|.$$

Stirling's approximation gives the asymptotic decay $\log P_{\text{perm}} \sim -n_V n_H \log n_V$ as $n_H \to \infty$ with n_V fixed — equivalently, $P_{\text{perm}} \sim n_V^{-n_V n_H}$. For any $n_V \geq 2$, the fraction of bijections failing C1 at the T-level decays exponentially in the hidden-sector capacity n_H. At modest physical scales — say $n_V = 10$, $n_H = 100$ — the fraction is $\sim 10^{-981}$; at the scales C3 invokes, it is astronomically smaller. The theorem is therefore non-vacuous on two independent grounds: the precondition excludes a structurally distinguished class of decoupled bijections (those whose V-dynamics have no h-dependence), and that class is a vanishing fraction of the configuration space as C3 is engaged. C3 (high-capacity hidden sector) and C1 (non-trivial coupling) are independent conditions in statement, but C3 is the *engine* of C1's genericity: enlarging n_H shrinks the measure of C1-violating bijections super-exponentially fast.

Relation to prior work. The P-indivisibility phenomenon is well-documented in the open-systems literature: reduced dynamics need not preserve positivity [6]; divisibility failure is equivalent to information backflow [4, 5]; process-tensor Markovianity provides an alternative characterization [7]; non-Markovian reduced dynamics arise in the slow-bath regime [8]. The Bylicka-Johansson-Acín equivalence between CP-divisibility and distinguishability monotonicity [9] narrows the gap between divisibility and non-Markovianity, but only for *invertible* reduced dynamics; the marginal $T(t)$ here is generally non-invertible, so the load-bearing property for the general argument remains P-indivisibility of the marginal process. Milz et al. [10] confirm CP-divisibility does not imply Markovianity in the process-tensor sense.

Continuous-time extension. The Hamiltonian flow on finite-dimensional phase space preserves Liouville measure on compact energy surfaces. $T_{ij}(t)$ is continuous with $T(0) = I$. By (C1), $T(t)$ departs from the permutation class for $t > 0$. By Poincaré recurrence, $\exists t_R$: the set of hidden states with $\pi_V(\varphi_{t_R}(x_i, h)) = x_i$ has measure $> 1 - \delta$ for any $\delta > 0$. For small δ, this gives non-monotonic trace distance, establishing P-indivisibility in continuous time.

The recurrence argument establishes P-indivisibility in principle. The following lemma shows that under (C2), information backflow occurs on observable timescales — not merely at Poincaré recurrence times.

Lemma (Accessible-timescale backflow). *Under (C1)–(C3) with $\tau_S \ll \tau_B$, the non-Markovian mutual information $I(X_{<t}; X_{>t} \mid X_t)$ is $\mathcal{O}(1)$ for observation windows $t \sim k\tau_S$ with $k\tau_S \ll \tau_B$.*

Proof. The coupling H_{int} transfers visible-sector information to the hidden sector at each interaction, at rate $\sim 1/\tau_S$. Between interactions, the hidden sector evolves under H_H with spectral gap $\Delta \sim 1/\tau_B$. The decay of correlations stored in the hidden sector is governed by $e^{-\Delta\tau_S}$. When $\tau_S \ll \tau_B$, $\Delta\tau_S \ll 1$, so the decay per visible-sector transition is $1 - e^{-\Delta\tau_S} \approx \Delta\tau_S \ll 1$: correlations survive each step essentially intact. After k transitions spanning a time $k\tau_S \ll \tau_B$, the cumulative decay is $e^{-k\Delta\tau_S} \approx 1 - k\Delta\tau_S$, which remains close to unity. The hidden sector therefore retains $\sim k$ bits of visible-sector history over this window, and the $(k+1)$-th transition reads back stored correlations through H_{int}, producing history-dependent transition probabilities — i.e., information backflow. More precisely, the mutual information satisfies:

$$I(X_{<t}; X_{>t} \mid X_t) \geq I_0(1 - k\Delta\tau_S) = I_0\left(1 - \frac{k\tau_S}{\tau_B}\right)$$

where $I_0 > 0$ is the single-step information transfer from (C1). For $k\tau_S \ll \tau_B$, this remains $\mathcal{O}(I_0)$ — comparable to the single-step coupling strength, not exponentially suppressed. The bound saturates (C3): the maximum storable history is $\log_2 m$ bits (§3.4), so persistent backflow over K transitions requires $m \geq 2^K$ when the per-transition contributions accumulate without hidden-state reuse (the independence hypothesis made explicit in §3.4) — amply satisfied when $N_H \gg N_V$. □

Role of (C2) and (C3). The Poincaré recurrence argument and the accessible-timescale lemma are independent: the former establishes P-indivisibility from (C1) and finiteness alone; the latter shows that (C2) and (C3) promote it from a formal property to an observationally dominant one.

Weak-coupling and fast-bath regimes. Known P-divisible systems with non-zero coupling illustrate the necessity of (C2), not a failure of the theorem. The Jaynes-Cummings model with Lorentzian spectral density is P-divisible when dissipation dominates coupling — the fast-bath regime ($\tau_B \ll \tau_S$), violating (C2). The Davies-Merkli Markovian convergence theorem likewise applies in the Born-Markov (fast-bath) limit. The §3.4 necessity proof for (C2) establishes this formally.

2.4 Explicit Construction: A Minimal Model

Visible sector: $x \in \{0, 1\}$. Hidden sector: $h \in \{1, \dots, 6\}$. Dynamics: the permutation

$$\sigma = (0{,}1 \leftrightarrow 1{,}1)\ (0{,}2 \leftrightarrow 1{,}2)\ (0{,}3 \leftrightarrow 0{,}4)\ (0{,}5 \leftrightarrow 0{,}6)\ (1{,}3 \leftrightarrow 1{,}4)\ (1{,}5 \leftrightarrow 1{,}6)$$

satisfying (C1)–(C3). Since $\sigma^2 = \mathrm{id}$: at $t = 1$, two of six hidden states flip x, giving

$$T(1,0) = \begin{pmatrix} 2/3 & 1/3 \\ 1/3 & 2/3 \end{pmatrix}$$

At $t = 2$ ($\sigma^2 = \mathrm{id}$): $T(2,0) = I$. A Markov chain would predict continued mixing: $T(1,0)^2 = \left(\begin{smallmatrix} 5/9 & 4/9 \\ 4/9 & 5/9 \end{smallmatrix}\right)$. The actual dynamics show complete *un-mixing*. Computing the intermediate propagator:

$$\Lambda(2,1) = T(2,0) \cdot [T(1,0)]^{-1} = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$$

Negative entries — no valid stochastic matrix exists. **P-indivisible.** □

Independent check (process-tensor non-Markovianity). The same model also fails the Pollock et al. [7] multi-time Markov condition without invoking $T(t)$-invertibility: $P(x_2 = 0 \mid x_1 = 1, x_0 = 0) = 1$ versus $P(x_2 = 0 \mid x_1 = 1) = 1/3$. Conditioning on x_0 changes the one-step prediction from 1/3 to 1 — a memory effect computed directly from multi-time conditional probabilities, valid regardless of $\det T(t)$. The non-Markovian character is therefore confirmed under both the divisibility-based criterion (Rivas et al. [4, 5]; Bylicka et al. [9] in the invertible case) and the process-tensor criterion (Pollock et al. [7] in the general case).

The mechanism. The hidden sector acts as a memory register. The transpositions $(0, h) \leftrightarrow (1, h)$ for $h \in \{1, 2\}$ flip x while preserving h, so a state at $x = 1$ with $h \in \{1, 2\}$ after step 1 carries the record "I was at $x = 0$." Step 2 reads this record and flips x back — information backflow impossible for a memoryless process. P-indivisibility is more extreme, not less, when memory is more persistent.

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3. THE EMERGENCE OF QUANTUM MECHANICS

3.1 The Stochastic-Quantum Correspondence

By Barandes' correspondence [11, 12], any P-indivisible stochastic process on a finite configuration space of size n embeds into a unitarily evolving quantum system on $\mathcal{H}$ of dimension $\leq n^3$. P-indivisible transition matrices develop negative entries at fine time resolution; the Stinespring dilation theorem guarantees a dilated Hilbert space and unitary $\tilde{U}(t)$ whose unistochastic matrix $|\tilde{U}(t)|^2$ marginalizes over the ancilla index to $T_{ij}(t)$ — transition probabilities are exactly Born-rule probabilities. The proof requires a finite-dimensional configuration space (Lemma 1); Calvo [13] extends the correspondence to infinite dimensions, but this is not required here. Pimenta [14] distinguishes P-divisible and P-indivisible stochastic-quantum dynamics: the correspondence is non-trivial only in the P-indivisible regime, which is precisely the regime produced by §2.3. An independent derivation using only Stinespring's dilation theorem [15] and standard partial-trace properties [16] is given in §3.2; either route alone suffices.

Scope of the correspondence (what it does and does not deliver). Because the entire emergence-of-QM claim rests on this external result, we state its scope precisely rather than treat it as a black-box equivalence. The correspondence is a published theorem (Barandes, *Philosophy of Physics* 3(1):8, 2025 [11]; proof in [12]) and is genuinely load-bearing here, but four scope conditions are carried throughout the framework and should be read as standing qualifications on every downstream "QM emerges" statement. (i) *Directionality and non-uniqueness.* The map is many-to-one in both directions; the complex matrix Θ with $T_{ij} = |\Theta_{ij}|^2$ is guaranteed to exist but is not unique (an entry-wise phase freedom analogous to a gauge potential), so the correspondence does not single out a unique quantum system from a stochastic process. (ii) *Scaffolding versus full QM.* The stochastic data directly supply only configuration-diagonal random variables; non-commuting observables enter as emergent patterns ("emergeables"), and spin enters by dilating the Hilbert space and imposing a rotation representation rather than being read off the stochastic side. The tensor-product structure of composite systems is on a different footing: for spatially separated visible subsystems the scope-of-equivalence remark in §3.4 derives the factorization $\mathcal{H}_{V_A} \otimes \mathcal{H}_{V_B}$ from the locality of the coupling graph (the spatial Markov property of range-1 dynamics), with entanglement arising from shared boundary history, rather than importing it on the Hilbert-space side. The bare correspondence therefore reconstructs the kinematic scaffolding and Born-rule transition statistics of quantum mechanics, not the full observable algebra for free; for the lattice substratum the composite-system factorization is in addition derived from locality, which leaves the spin representation as the one component still supplied by the dilation rather than read off the substratum. Its unification with the staggered-to-Dirac spin structure of the lattice substratum — whose spinor decomposition is keyed to the lattice's bipartite partition — is the open step. One premise of that step is secured at the linear level: the bipartite grading itself survives observation exactly — the effective kernel obeys $\Gamma_V K_{\text{eff}}(E)\,\Gamma_V = -K_{\text{eff}}(-E)$ for any site partition ([SM §4.8]), so the marginalization cannot generate a grading-breaking (staggered-mass) term and the structure the spinor decomposition is keyed to reaches the observer intact. (iii) *Restricted process class.* The representation $T_{ij} = |\Theta_{ij}|^2$ imposes substantive structure on the admissible processes; in particular Doukas (2026) shows the differentiable form forces the first-order rate matrix to vanish, excluding ordinary continuous-time Markov chains — so "any stochastic process" is narrower than it sounds, and the qualifying regime is precisely the indivisible one §2.3 produces, not the generic one. The restriction also runs the other way. Doukas's accompanying bound — that marginalization over a *finite* environment produces no first-order dissipative term, secular leakage requiring an infinite reservoir — is, in continuous-time form, the same structural fact as the second-order estimate of §3.2 for the discrete substratum: a finite deterministic substratum observed by marginalization cannot generate first-order leakage, and its emergent dynamics is thereby confined to the near-unitary corner of process space, displaced from exact unitarity only by the residual dissipation $\mathcal{D}$. What appears as a restriction on the correspondence's process class is, under the present architecture, the mechanism's signature: the framework occupies the corner Doukas identifies not by selection but by structural necessity, conditional on its finiteness commitment (A1, argued at the axiom layer). This is explanatory unification rather than corroboration — near-unitarity is an empirical input elsewhere in the framework — but it converts scope condition (iii) from a qualification into a prediction: fundamental dynamics of continuous-time Markov type would require an actually infinite hidden sector, and the finite-ϵ displacement $\mathcal{D} \sim 10^{-64}$ is the corner-occupancy claim's falsifiable residue. (iv) *Reception.* The result is recent and located at the physics/philosophy boundary. Its mathematical core has since been independently reworked and generalized: the lift structure and its divisibility mechanics by Doukas [22], the countable-dimension extension by Calvo [13], the divisible/indivisible classification by Pimenta [14], and the Tsirelson-bound result in a mainstream physics venue [24, forthcoming in Phys. Rev. A]. The strongest interpretive readings (deflation of the wave function, of entanglement) remain contested philosophical overlays on the mathematical core, which is all the framework requires. None of (i)–(iv) undercuts the framework's use of the correspondence to obtain Born-rule statistics and unitary evolution from the §2.3 process; they bound what may be claimed to follow *from* it.

Independent corroboration of the trace-out mechanism. The technical move from a deterministic substrate to non-Markovian visible dynamics via the trace-out of inaccessible degrees of freedom — central to both routes — is now established at theorem level in the open systems literature. Brandner [17, 18] proves that for autonomous linear evolution equations, integrating out inaccessible degrees of freedom (whether external environments or internal subsystems) yields well-defined non-Markovian visible dynamics in a controlled weak-memory regime, with explicit error bounds and a convergent perturbation scheme. This is the mathematical apparatus the present construction relies upon, derived independently for general open systems. Direct experimental demonstrations of the mechanism include Mehl et al. [19], who showed that hidden slow degrees of freedom in a colloidal system produce non-Markovian visible dynamics violating naive Markovian fluctuation theorems, and Gröblacher et al. [20], who observed non-Markovian Brownian motion in a macroscopic micromechanical oscillator. On the quantum side, Kim [21] shows that monitored quantum systems are formally quantum hidden Markov models, with a rigorous correspondence to classical hidden Markov models for setups involving Haar-random unitaries and measurements — the same formal structure as the stochastic-quantum bridge. The framework's central technical move is therefore standard rather than speculative.

From phase space to configuration space. The transition probabilities T_{ij} are projections of the full phase-space flow onto the configuration sector of Γ_V, with momenta and hidden-sector degrees of freedom absorbed into the Liouville marginalization. The resulting process lives on the discrete configuration space of Lemma 1.

Continuous spectra and the finite-substratum commitment. Lemma 1 fixes the visible configuration space as finite, and the Barandes correspondence as applied here produces an emergent Hilbert space of finite dimension. Standard quantum mechanics makes heavy use of continuous-spectrum observables — position on $\mathbb{R}^3$, momentum, continuous-spectrum scattering — which strictly require infinite-dimensional Hilbert spaces and unbounded self-adjoint operators. The framework's position on this is that continuous spectra are the infrared limit of the discrete spectrum produced by Lemma 1, not a deeper reality that the framework fails to capture. The quantitative separation is extreme: for a visible sector bounded by the cosmological horizon with discreteness scale $\epsilon = 2l_p$, the number of distinguishable positions is $|\mathcal{C}_V| \sim (L_H/\epsilon)^3 \sim 10^{183}$, and the holographic bound on emergent Hilbert-space dimension is $e^{S_{\rm dS}} = e^{10^{122}}$. The discretization error separating this finite-dimensional structure from a continuum approximation is $\lesssim 1/\sqrt{\dim \mathcal{H}} \sim e^{-10^{121}}$ per eigenvalue — operationally zero at any accessible resolution. Deviations from continuum QM at scale E are suppressed by $(E/M_{\rm Pl})^2$, the same scaling that controls Lorentz-invariance violation and trans-Planckian corrections; no independent continuous-spectrum prediction is needed or claimed. Calvo [13] extends the Barandes correspondence to countable-infinite dimensions for completeness, but the finite-dim correspondence is the load-bearing one for the framework — every result in §2–§3 runs on Lemma 1 directly.

Three features emerge. The Schrödinger equation arises from the differentiability of $U(t)$. The Born rule is the equilibrium distribution of the indivisible process [11, 12]. The action scale $\hbar$ enters when defining $\hat{H}(t) \equiv i\hbar\, \partial_t U \cdot U^\dagger$, converting dimensionless rates to energy units; its value requires additional physical input from the partition.

Phase uniqueness from continuous-time data. The relation $T_{ij}(t) =$

$|U_{ij}(t)|^2$ discards phase information at any single time. Doukas [22] identifies this gap for discrete-time data. The resolution: continuous-time data $\{T_{ij}(t)\}_{t\in\mathbb{R}}$ provides strictly more information.

Lemma (phase-locking). *Let H be Hermitian on $\mathbb{C}^n$ with eigenbasis $\{|k\rangle\}$, eigenvalues $\{E_k\}$, and $V_{ik} = \langle i|k\rangle$. Assume: (G1) non-degenerate spectrum; (G2) non-degenerate energy gaps; (G3) $V_{ik} \neq 0$ for all i, k. Then $T_{ij}(t) = |\langle i|e^{-iHt}|j\rangle|^2$ for all i, j, t uniquely determines H up to an overall energy shift and basis phase conventions.*

Proof. Write:

$$T_{ij}(t) = \sum_{k,l} V_{ik}\, V_{jk}^*\, V_{jl}\, V_{il}^*\; e^{-i(E_k - E_l)t}$$

By (G2), the frequencies $\omega_{kl} = E_k - E_l$ are distinct for distinct pairs with $k \neq l$. Fourier-transforming yields:

$$a_{ij}^{kl} = V_{ik}\, V_{jk}^*\, V_{jl}\, V_{il}^*$$

The moduli $|V_{ik}|$ follow from $a_{ii}^{kl} = |V_{ik}|^2 |V_{il}|^2$ (non-zero by (G3)): the rank-1 outer-product structure determines $|V_{ik}|^2$ up to a global scale, and unitarity $\sum_k |V_{ik}|^2 = 1$ fixes the scale. The phases carry a double-difference structure:

$$\arg(a_{ij}^{kl}) = (\varphi_{ik} - \varphi_{il}) - (\varphi_{jk} - \varphi_{jl})$$

The gauge freedom preserving all double differences is $\varphi_{ik} \to \varphi_{ik} + \alpha_i + \beta_k$ (physically irrelevant basis rephasing). Fixing $\varphi_{1k} = 0$, $\varphi_{i1} = 0$, each remaining phase is $\varphi_{ik} = \arg(a_{i1}^{k1})$, directly extracted from Fourier data. $H = V\,\mathrm{diag}(E_k)\,V^\dagger$ is unique up to energy shift and phase conventions. □

Scope of the lemma under the ancilla-marginal form, and its extension. The lemma as stated assumes the strict form $T_{ij}(t) = |\langle i|e^{-iHt}|j\rangle|^2$ on an n-dimensional space — the trivial-ancilla case of the definition in §3.4. The emergent object is the ancilla-marginal (§3.1), and the lemma extends to it in a stronger form. Writing $U(t) = e^{-i\hat{H}t}$ on the dilated space of dimension $D = n\, m_a$ with eigenbasis $\{|a\rangle\}$ and configuration projectors P_i, the marginal takes the exact form

$$T_{ij}(t) = \frac{1}{m_a} \sum_{a,b} e^{-i(E_a - E_b)t}\, \langle a|P_i|b\rangle\, \langle b|P_j|a\rangle,$$

so under non-degenerate gaps of the *dilated* spectrum, Fourier analysis of the continuous-time visible data recovers, for each eigenvector pair (a, b), the rank-one coefficient matrix $A_{ij}^{ab} = \frac{1}{m_a}\langle a|P_i|b\rangle\langle b|P_j|a\rangle$, hence the coherence vector

$\langle a|P_i|b\rangle$ up to one phase per pair. The residual gauge is larger than the diagonal rephasing of the original lemma — one phase per pair, $D(D-1)/2$ parameters, under which $T(t)$ is exactly invariant — but the requirement that the reconstructed P_i be orthogonal projectors summing to the identity is a nonlinear constraint tying the pairs together, and it eliminates precisely the excess. The reduction can be shown directly. Writing a per-pair rephasing as the linearized perturbation $M_i[a,b] \mapsto M_i[a,b](1+i\theta_{ab})$ with θ antisymmetric (where $M_i[a,b] = \langle a|P_i|b\rangle$), the linearized idempotency constraint $\delta(M_i^2) = \delta(M_i)$ reads $\sum_b(\theta_{ab}+\theta_{bc})\, M_i[a,b]\, M_i[b,c] = \theta_{ac}\, M_i[a,c]$ for all i,a,c. The diagonal coboundary $\theta_{ab} = \alpha_a - \alpha_b$ satisfies this identically: its left side telescopes to $(\alpha_a - \alpha_c)\sum_b M_i[a,b]M_i[b,c] = (\alpha_a-\alpha_c)\, M_i^2[a,c] = (\alpha_a-\alpha_c)\, M_i[a,c]$ using $P_i^2 = P_i$, which is the right side. This exhibits the $D-1$ diagonal-gauge directions (the D shifts α_a modulo a global constant that acts trivially) as an analytic null space of the constraint, for every projector family. That these are the *only* null directions — that the constraint admits no pair-phase deformation beyond the coboundary — is verified numerically: the linearized idempotency operator has null space of dimension exactly $D-1$ for generic $\hat{H}$ across $(n, m_a) = (3,4), (2,3), (4,2), (3,3), (2,4)$, and the constructive Fourier extraction has been demonstrated end-to-end. The marginal data therefore determine — generically, and at least locally — the *dilated* Hamiltonian and the configuration projectors up to an energy shift and diagonal phase conventions; the reconstruction reaches the dilation itself, not only an n-dimensional effective generator. The status of each component: the Fourier structure, its rank-one property, and the coboundary null directions are exact algebra; that the coboundary *exhausts* the null space is established numerically and awaits a written completeness proof (the natural route is that generic projector families make the constraint operator's kernel no larger than the coboundary, a genericity statement); the genericity conditions (G1)–(G3) transfer to the dilated spectrum, gaps, and coherence vectors. One caveat applies in sharpened form: the idealized permutation dynamics has dilated eigenphases at roots of unity — equally spaced, hence maximally gap-degenerate — so the lemma's genericity holds for the physical continuous-time dynamics away from that idealized limit, not at it; this is the same idealization boundary carried by the finite-ϵ displacement $\mathcal{D}$.

Conditions (G1)–(G3) fail only on a measure-zero set of Hamiltonians. A stronger argument applies whenever the partition itself breaks symmetries of $H_{\rm tot}$. The genericity conditions concern the *effective* visible-sector Hamiltonian $\hat{H}_{\rm eff}$ — the operator emerging after the trace-out — not the total Hamiltonian $H_{\rm tot}$. The relevant distinction is whether a symmetry of $H_{\rm tot}$ *preserves the visible/hidden factorization* or not. A symmetry that fails to preserve the factorization — which includes the spatial symmetries an observer-centered partition breaks, since a partition centered on a specific observer maps boundary visible sites to hidden sites under translation, rotation about distant points, and boost — has no well-defined action on the visible factor alone, and the degeneracies it protects in $H_{\rm tot}$ are lifted in $\hat{H}_{\rm eff}$. (By contrast, a symmetry that *does* factorize as $S_V \otimes S_H$ descends to the visible symmetry S_V of $\hat{H}_{\text{eff}}$, because the uniform-bath partial trace is invariant under any unitary on the hidden factor; such factorization-preserving symmetries are not lifted, and the visible-internal ones among them — those acting entirely within the visible sector and commuting with the trace-out — are precisely the gauge symmetries of the emergent description. Gauge degeneracies correspond to physically equivalent states and do not affect the phase-locking argument, which operates on gauge-inequivalent transition data.) Because the spatial symmetries of an observer-centered partition are factorization-breaking, $\hat{H}_{\text{eff}}$ satisfies G1 (non-degenerate spectrum) not by typicality in an abstract ensemble but because the partition acts as a symmetry-breaking mechanism. Condition G2 (non-degenerate energy gaps) is not symmetry-protected — gap coincidences are a codimension-one condition rather than a group-theoretic one — so it is not delivered by the partition mechanism and rests on genericity directly. G2 holds for generic $\hat{H}_{\text{eff}}$; it can fail only on a measure-zero set, and we are not aware of a structural feature of the substratum that forces such failure systematically. (In particular, single-cycle hidden dynamics does not: although the unitary cycle operator has equally-spaced phases on the unit circle, the Hermitian effective generator $\hat{H}_{\text{eff}}$ has real eigenvalues that are not equally spaced, and retains non-degenerate gaps after the trace-out.)

3.2 Independent Derivation via Stinespring Dilation

An independent derivation of the emergent quantum description uses only Stinespring's dilation theorem [15] and standard properties of the partial trace [16], without invoking the stochastic-quantum correspondence of §3.1. The two routes are independent in their inputs and in the results they invoke (the correspondence's own proof also proceeds via Stinespring-type dilation, so the shared bedrock is Stinespring's 1955 theorem); either alone suffices.

Setup. The finite configuration spaces $\mathcal{C}_V = \{x_1, \dots, x_n\}$ and $\mathcal{C}_H = \{h_1, \dots, h_m\}$ (Lemma 1) embed into Hilbert spaces $\mathcal{H}_V = \mathbb{C}^n$ and $\mathcal{H}_H = \mathbb{C}^m$ via $|i\rangle \leftrightarrow x_i$ and $|k\rangle \leftrightarrow h_k$. This introduces no quantum postulates: it is the canonical identification of probability distributions on a finite set with diagonal density matrices.

Lemma (permutation unitarity). *Any bijection $\varphi : \mathcal{C}_V \times \mathcal{C}_H \to \mathcal{C}_V \times \mathcal{C}_H$ defines a unitary U_φ on $\mathcal{H} = \mathcal{H}_V \otimes \mathcal{H}_H$.*

Proof. Define $U_\varphi|i,k\rangle = |\varphi(x_i, h_k)\rangle$. Since φ is a bijection, U_φ permutes the orthonormal basis, hence is unitary. □

A finite permutation does not by itself generate a continuous flow: on a finite discrete configuration space the dynamics is the discrete group $\{U_\varphi^k\}$, and any continuous one-parameter interpolation is *additional structure*, chosen rather than derived. When such an interpolation is introduced — a strongly continuous one-parameter unitary group U_t with $U_1 = U_\varphi$, which always exists but is not unique (any branch of $\log U_\varphi$, differing by 2π shifts of eigenphases, defines one) —

Stone's theorem yields $U_t = e^{-i\hat{H}t}$ for Hermitian $\hat{H}$; the framework's continuous-time statements are conditional on that choice, with the branch ambiguity part of the phase-retrieval gauge catalogued below.

Lemma (Reverse direction, scoped). *Any single one-step doubly stochastic matrix on a finite configuration space $\mathcal{C}_V$ can be realised as the one-step marginal of a deterministic bijection on $\mathcal{C}_V \times \mathcal{C}_H$ with uniform prior on $\mathcal{C}_H$, for some finite $\mathcal{C}_H$ — exactly when the transition probabilities are rational, and to arbitrary precision otherwise.* The restriction to doubly stochastic matrices is forced, not elective: a uniform-prior bijection marginal preserves the uniform distribution and is therefore doubly stochastic, so a merely row-stochastic matrix — one sending both states of a two-state space to the first, say — is not realisable in this form. The emergent statistics this converse serves lie inside the realisable class: ancilla-uniform marginals of unistochastic matrices are doubly stochastic. The scope is the matrix, not the process: realising prescribed *multi-time* statistics — a family of joint distributions over trajectories — with a single bijection and a single uniform prior is a stronger demand and is not claimed by this lemma; where multi-time structure is used below, it enters through the dilation actually constructed, not through this converse.

Proof. Any marginal of a bijection with uniform prior is a doubly stochastic matrix, and by Birkhoff–von Neumann is a convex combination of permutation matrices. A bijection on $\mathcal{C}_V \times \mathcal{C}_H$ realising the required mixture is obtained by letting $\mathcal{C}_H$ enumerate the permutations; for multi-step processes, the construction extends by history dilation. □

Remark. This is the correspondence at the stochastic-process level: bijection $\leftrightarrow$ stochastic matrix via marginalization, both directions. It is *not* a full CPTP-channel correspondence — channels that create coherences from diagonal inputs have no permutation-unitary realisation with uniform ancilla. For the characterization theorem (§3.4), which is stated in terms of transition probabilities, this is the appropriate scope.

The quantum channel. The observer's ignorance of the hidden sector (Lemma 3) corresponds to $\rho_H = I_m/m$. The visible-sector quantum channel is

$$\Phi(\rho_V) = \mathrm{Tr}_H\left[U_\varphi\,(\rho_V \otimes \rho_H)\,U_\varphi^\dagger\right]$$

This is CPTP by a standard result [16, Theorem 8.1], with Kraus representation $\Phi(\rho_V) = \sum_{k,l} K_{kl}\,\rho_V\,K_{kl}^\dagger$ where $K_{kl} = m^{-1/2}\langle l|U_\varphi|k\rangle_H$. The triple $(\mathcal{H}_H, U_\varphi, \rho_H)$ is the Stinespring dilation [15] of Φ.

Theorem (Born rule recovery). *The classical transition probabilities T_{ij} (§1.4) equal the Born-rule probabilities of Φ.*

Proof. $P(j|i) = \langle j|\Phi(|i\rangle\langle i|)|j\rangle = m^{-1}\sum_{k,l} |\langle j,l|U_\varphi|i,k\rangle|^2$. Since U_φ is a permutation, $\langle j,l|U_\varphi|i,k\rangle = \delta_{(j,l),\varphi(i,k)}$. Thus $P(j|i) = m^{-1}\sum_k \delta_{j,\pi_V(\varphi(x_i,h_k))} = T_{ij}$. □

Remark (failure of (C1)). Condition (C1) does not suffice. For $|V| = |H| = 2$ and $\varphi(x,h) = (h,x)$ one has $T_{ij} = 1/2$ for all i,j, yet $\Phi(\rho) = I/2$ identically: a constant channel, hence measure-and-prepare. Exhaustive enumeration gives the same verdict for every (C1)-satisfying bijection at $|V| = |H| = 2$, and for 612 of 648 at $|V| = 2$, $|H| = 3$ (verification code: `papers/oi_lattice_code/coherence/`). The obstruction is structural: (C1) constrains the diagonal of Φ, whereas entanglement-breaking is a property of $\mathrm{id} \otimes \Phi$ and is not fixed by the action on single-system inputs.

What replaces (C1) is not a further condition on T but a condition on the partition, and it is established for a specific realization rather than for arbitrary bijections. Take the substratum of §1.2 on a lattice Λ with $q = 2$, state $(u,v) \in \mathbb{F}_2^\Lambda \times \mathbb{F}_2^\Lambda$ and the reversible nearest-neighbour update

$$u'_x = \sum_{z \sim x} u_z + v_x, \qquad v'_x = u_x,$$

the visible sector being the sites of a region $R \subseteq \Lambda$. Write M_{HV} and M_{VH} for the blocks of this update carrying visible input to hidden output and hidden input to visible output, $\kappa = \dim\ker M_{HV}$, $w = \operatorname{rank} M_{VH}$, and $s = \dim V = 2|R|$. The inner and outer boundaries of R are $\partial^- R = \{x \in R : \exists\, z \notin R,\ z \sim x\}$ and $\partial^+ R = \{y \notin R : \exists\, z \in R,\ z \sim y\}$.

Lemma 1 (boundary bound). $\operatorname{rank} M_{HV} \le |\partial^+ R|$ *and* $\operatorname{rank} M_{VH} \le |\partial^- R|$.

Proof. For $y \notin R$ the component $v'_y = u_y$ is a hidden input, and $u'_y = \sum_{z\sim y} u_z + v_y$ involves a visible input only if some $z \sim y$ lies in R, that is only if $y \in \partial^+ R$; so M_{HV} has non-zero rows only among $\{u'_y : y \in \partial^+ R\}$. Dually, for $x \in R$ the component $v'_x = u_x$ is a visible input, and u'_x involves a hidden input only if $x \in \partial^- R$. □

One update step transports information one lattice site, so the sectors resolve one another only across the cut. This argument is a statement about supports and uses no property of the coefficient ring: it holds verbatim over $\mathbb{Z}_q$, and for a general nearest-neighbour bijection it yields the weaker conclusion that the hidden output depends on the visible input only through $\partial^- R$.

Lemma 2 (channel normal form). *Write $A = M_{VV}$, $B = M_{VH}$, $C = M_{HV}$ for the blocks of the update, and L_{RR} for the adjacency of Λ restricted to R. Then: (i) A is invertible, with $A^{-1}(a_u, a_v) = (a_v,\ a_u - L_{RR}\, a_v)$, so the visible block alone induces the permutation (Clifford) unitary $V_A|i\rangle = |Ai\rangle$; (ii) $\Phi(|i\rangle\langle i'|)$ vanishes unless $C(i - i') = 0$; (iii) on the Weyl operators the channel acts, with no phase, as*

$$\Phi(X^a Z^b) = [\, Ca = 0\,]\, [\, B^T A^{-T} b = 0\,]\; X^{Aa}\, Z^{A^{-T} b},$$

so that $\Phi = \mathrm{Ad}_{V_A} \circ \Phi_G$, where $\Phi_G = |G|^{-1} \sum_{g \in G} g \cdot g^\dagger$ is the uniform mixture *over the group G of Weyl operators $X^\alpha Z^\beta$ with $\alpha \in \operatorname{im}(A^{-1}B)$ (translations) and $\beta \in \operatorname{im} C^T = (\ker M_{HV})^\perp$ (phases); (iv) G is abelian, of order $2^{\,s-\kappa+w}$.*

Proof. (i) is the displayed inverse, read off from $v'_x = u_x$ and $u'_x = (L_{RR}u)_x + v_x$ on R; invertibility of A is what makes $i \mapsto Ai$ a relabelling. (ii) The hidden outputs of i and i' agree for some hidden input exactly when $C(i-i')=0$, and then for every hidden input. (iii) With the hidden input uniform, $\Phi(|i\rangle\langle i'|) = [\,C(i-i') = 0\,]\ \mathbb{E}_{t\in\operatorname{im} B}\,|Ai+t\rangle\langle Ai'+t|$. Summing this against $X^aZ^b = \sum_i (-1)^{b\cdot i}\,|i+a\rangle\langle i|$ and substituting $i \mapsto i - A^{-1}t$ gives the displayed rule: the character sum over $\operatorname{im} B$ produces the second indicator and cancels every phase. The right-hand side is Ad_{V_A} applied to the Hilbert–Schmidt-orthogonal projection onto $\operatorname{span}\{X^aZ^b : Ca = 0,\ B^TA^{-T}b = 0\}$, and the uniform mixture over the symplectic complement of that index set — which is exactly G — is that projection. (iv) An element $X^\alpha Z^\beta$ with $\alpha \in \operatorname{im}(A^{-1}B)$, $\beta \in \operatorname{im} C^T$ commutes with another iff the cross-pairings $\alpha\cdot\beta'$ vanish; over all of G this is the single condition $CA^{-1}B = 0$, which holds identically for this update: B feeds hidden u into the visible u'-components, A^{-1} carries the u'-subspace onto the v-subspace, and C reads only the visible u-components. The translation and phase families intersect only in the identity, so $|G| = 2^{\operatorname{rank} B}\cdot 2^{\operatorname{rank} C} = 2^{\,w+(s-\kappa)}$. □

Theorem (separability threshold). *Let G be an abelian subgroup of the Weyl group on s qubits, of order 2^t modulo phases. Then $\Phi_G = |G|^{-1}\sum_{g\in G} g\,\cdot\,g^\dagger$ is entanglement-breaking if and only if $t = s$.*

Proof. Modulo phases the Weyl operators form the symplectic space $\mathbb{F}_2^{2s}$, on which abelian subgroups are exactly the isotropic subspaces; isotropy bounds $t \le s$. Any isotropic subspace of dimension t is carried to $\langle Z_1,\dots,Z_t\rangle$ by a symplectic transformation (Witt's extension theorem for the symplectic space $\mathbb{F}_2^{2s}$), and every symplectic transformation is implemented by a Clifford unitary U. Hence $U\Phi_GU^\dagger$ is complete dephasing on t qubits tensored with the identity on the remaining $s-t$. Complete dephasing is measure-and-prepare and therefore entanglement-breaking, and a tensor product of channels is entanglement-breaking exactly when every factor is — separability of the joint Choi state survives the partial trace onto any single factor pair, so one entangled factor Choi rules out separability of the whole; the identity on a non-trivial factor is not entanglement-breaking. Since conjugation by a unitary preserves the property, Φ_G is entanglement-breaking precisely when $s-t=0$. □

Corollary 1. *For one step of the displayed update on $\mathbb{F}_2$ — $u'_x = \sum_{z\sim x} u_z + v_x$, $v'_x = u_x$, with the hidden sector initially uncorrelated and maximally mixed — the visible channel Φ admits the Clifford-conjugated abelian Weyl-mixture normal form of Lemma 2 and is not entanglement-breaking if and only if $w < \kappa$.* Immediate from Lemma 2 and the Theorem with $t = s-\kappa+w$: entanglement breaking is invariant under composition with the unitary Ad_{V_A}, so Φ is entanglement-breaking exactly when Φ_G is, and $t=s$ reads $w=\kappa$ (Lemma 2 gives $w\le\kappa$, matching $t\le s$).

Corollary 2. *If $|\partial^- R| + |\partial^+ R| < 2|R|$ then Φ is not entanglement-breaking.*

Proof. $s = 2|R|$, so Lemma 1 gives $w + (s - \kappa) \leq |\partial^- R| + |\partial^+ R| < s$, that is $w < \kappa$. □

Remark. For a d-dimensional region of linear size ℓ, $|\partial R| = O(\ell^{d-1})$ against $2|R| = 2\ell^d$, so the hypothesis of Corollary 2 holds whenever $\ell > 2d$; in $d = 3$ this is roughly six sites across. Within this single-step result the criterion carries no dependence on $|H|$, since the coupling reads only the boundary; the counterexample above is the degenerate case, every visible site lying on the cut so that the region has no bulk. Whether the same holds over many steps is not settled here: repeated interaction can propagate boundary information into the hidden bulk, and a multi-time statement would require a process-tensor argument rather than a single-channel one. The suggestion that coherence is a bulk quantity and decoherence an area quantity — the accounting that reappears in the horizon entropy of §7 — is at this stage an interpretation of the one-step result, not a theorem about the asymptotic dynamics.

Remark (scope). Lemma 1 holds over any $\mathbb{Z}_q$ and, in weakened form, for arbitrary nearest-neighbour bijections. Lemma 2, the Theorem and both Corollaries are stated for $q = 2$; the symplectic and Clifford arguments carry over to prime q with $2^\bullet$ replaced by $q^\bullet$, while composite q requires module-theoretic treatment of rank and kernel and is not covered. All of this concerns the realization displayed above and is not a property of arbitrary finite bijections. For a general linear bijection the abelian property of Lemma 2(iv) can fail ($CA^{-1}B \neq 0$); the mixture is then over a non-abelian Weyl family, and entanglement breaking is governed by the isotropy of the surviving index set rather than by its order alone — not treated here. Two boundary markers on interpretation. Coherence preservation is a property of this realization, not an additional structural condition: (C1)–(C3) constrain coupling, timescales, and capacity, and neither contain nor imply a coherence requirement — the swap example satisfies (C1) while its channel is entanglement-breaking, and which realizations preserve one-step coherence is the geometric question answered above for the displayed update alone. And non-entanglement-breaking is a statement about a single channel, not about quantum mechanics: it supplies neither the observable algebra, nor preparations and interventions, nor tensor composition, nor multi-time statistics — the operational reconstruction claims rest on the dilation and correspondence arguments, not on Corollary 1 — and the entanglement-breaking status of one step does not determine whether the visible multi-time process is P-divisible: the hidden sector can carry memory even when a step's channel is entanglement-breaking, as a two-state system with a two-state hidden register already shows. The enumeration and the boundary bounds are reproduced by the deterministic tests in `papers/oi_lattice_code/coherence/` (`coherence_enumeration.py`); the transfer rule and normal form of Lemma 2, the threshold Theorem exhaustively at $s = 2, 3$, and both directions of Corollary 1 — a negative partial transpose wherever $w < \kappa$, an explicit measure-and-prepare form wherever $w = \kappa$ — are verified for $q = 2$ and $q = 3$ by `coherence_theorem_tests.py` in the same directory; the strictness of the dephasing/entanglement-breaking containment and the compatibility of an entanglement-breaking step with a P-indivisible

visible process are certified by `eb_pindivisibility_tests.py` there as well.

Theorem (CP-indivisibility). *The P-indivisibility of §2.3 implies CP-indivisibility of $\{\Phi_t\}$: there exist $t_2 > t_1 > 0$ with no CPTP map Λ satisfying $\Phi_{t_2} = \Lambda \circ \Phi_{t_1}$.*

Proof. CP-divisibility restricted to diagonal inputs reduces to P-divisibility. The contrapositive gives the result. □

By the Breuer-Laine-Piilo criterion [3, 4], CP-indivisibility implies non-monotonic trace distance (information backflow) — the quantum signature of non-Markovianity.

Approximate unitarity. On observable timescales $t \ll \tau_B$, the hidden-sector state is approximately frozen (conditions (C2)–(C3)). At $t = 0$ the channel generator is exactly Hamiltonian, $d\Phi_t/dt|_{t=0}(\rho_V) = -i[\hat{H}_{\text{eff}}, \rho_V]$ with $\hat{H}_{\text{eff}} = H_V + \mathrm{Tr}_H[V_{\text{int}}(\mathbb{1} \otimes \rho_H)]$ the bath-averaged coupling; the dissipative part of the generator vanishes at $t = 0$. At finite time the visible channel acquires two distinct dissipative contributions, which it is important to separate. (i) Even in the strictly frozen-bath limit ($\tau_B \to \infty$) the channel is not unitary: it is a mixture over the bath ensemble of bath-conditioned visible unitaries $e^{-i(H_V + \langle h|V_{\text{int}}|h\rangle)t}$, which dephases by the *spread* of those conditioned Hamiltonians — a contribution of order $\|V_{\text{int}}\|^2$ (the coupling strength), independent of τ_B. (ii) The correction from the bath's actual motion away from the frozen limit enters at order $(\tau_S/\tau_B)^2$. Consequently the dynamics is approximately the Schrödinger equation generated by $\hat{H}_{\text{eff}}$ to the extent that *both* the slow-bath condition $\tau_S \ll \tau_B$ *and* a weak-coupling condition (small conditioned-Hamiltonian spread, $\|V_{\text{int}}\|\,\tau_S \ll 1$) hold; of the two contributions, the bath-motion term is fixed by the timescale hierarchy at $\mathcal{O}((\tau_S/\tau_B)^2) \sim 10^{-64}$, while the conditioned-Hamiltonian spread is not — its magnitude for the cosmological partition awaits the substratum interaction structure. Where these hold, the phase-locking lemma (§3.1) determines $\hat{H}_{\text{eff}}$ uniquely from continuous-time transition data, up to the residual diagonal-unitary rephasing freedom that is physically trivial (basis convention); the residual dissipation $\mathcal{D}$, if measurable at sufficient precision, is a prediction of the framework rather than an approximation artifact.

Comparison of routes.

	Barandes route (§3.1)	Stinespring route (§3.2)
Input	P-indivisible process on $\mathcal{C}_V$	Bijection on $\mathcal{C}_V \times \mathcal{C}_H$ + marginalization
Bridge	Stochastic-quantum correspondence [11, 12]	Stinespring dilation [15] + partial trace [16]
Output	Unitary with $T_{ij} = \lvert U_{ij}\rvert^2$	CPTP channel with $T_{ij} = \langle j\lvert\Phi(\lvert i\rangle\langle i\rvert)\rvert j\rangle$

	Barandes route (§3.1)	Stinespring route (§3.2)
Born rule	Equilibrium of indivisible process	Partial trace structure (above)
Scope	Any P-indivisible process	Processes from marginalized bijections

The Barandes route is more general (any P-indivisible process); the Stinespring route requires only textbook results [15, 16] and delivers additional structure: the tensor product $\mathcal{H}_V \otimes \mathcal{H}_H$, genuine quantum coherence, CP-indivisibility, and approximate unitarity: the leading departure from the frozen-limit description enters at $\mathcal{O}(\tau_S/\tau_B) \sim 10^{-32}$ as an absorbable Hermitian drift of $\hat{H}_{\text{eff}}$, with genuine non-unitarity $\mathcal{D}$ entering through the conditioned-Hamiltonian spread and the second-order bath-motion term $\mathcal{O}((\tau_S/\tau_B)^2) \sim 10^{-64}$ (§3.2). The first-order drift is moreover common-mode: the constraint coupling (C1) acts on total energy–momentum, so the slow change of the traced-out horizon state rescales $\hat{H}_{\text{eff}}$ uniformly and cancels in all dimensionless ratios — which is why the 10^{-32} figure is absorbable rather than observable — with the composition-differential remainder suppressed by the tidal factor $(L_{\text{lab}}/R_H)^2 \sim 10^{-52}$ and independently capped by equivalence-principle tests. The two routes agree on all observables — both produce the same $T_{ij}(t)$ — establishing that the emergence of QM is robust and overdetermined. The Barandes route powers (ii) $\implies$ (i) in the characterization theorem; the Stinespring route supplies (iii) $\implies$ (i) independently.

Dependency scope of the correspondence. The division of labor between the routes fixes the framework's exposure to the correspondence precisely: the sufficiency direction — embedded observation under (C1)–(C3) yields the emergent quantum description, with its Born-rule statistics, Hamiltonian uniqueness (phase-locking, whose proof is self-contained Fourier analysis), and every downstream quantitative result of the framework, including the $\hbar$ relation derived from the cosmological-horizon partition — is established by the Stinespring route together with the reconstruction-theorem necessity statement below, and survives without the correspondence entirely. The correspondence [11, 12] is load-bearing for one result only: the necessity direction of the characterization theorem (§3.4), where the QM $\Leftrightarrow$ P-indivisibility biconditional powers (ii) $\implies$ (i). A failure of the correspondence would therefore leave the framework's constructive claims intact and would remove only the converse — that every quantum system requires an embedded-observation realization.

An independent necessity statement. Both routes above are constructive: they exhibit quantum mechanics as the description of the marginalized process. The operational reconstruction theorems [Hardy, arXiv:quant-ph/0101012 (2001); Masanes and Müller, New J. Phys. **13**, 063001 (2011); Chiribella, D'Ariano and Perinotti, Phys. Rev. A **84**, 012311 (2011)] supply the converse direction from a disjoint axiom base: among finite-dimensional generalized probabilistic theories, quantum theory is the unique one satisfying a short list of operational requirements — tomographic locality, continuous reversibility, and, in the informational axiomatization, purification — stated without reference to a substratum, determinism, or a hidden sector. The requirements without the continuity clause admit exactly two solutions, classical and quantum probability theory; the P-indivisibility theorem of §2.3 — the framework's own contribution — is precisely the certificate that the marginalized process occupies the non-classical branch. The remaining load-bearing hypotheses each have an independent origin in the present construction rather than being read off the reconstructed output: finite dimensionality is Lemma 1; purification is exhibited rather than postulated, every visible mixed state arising as the marginal of the deterministic dilation (§3.2); no-signaling follows from the marginalization structure (§3.3); continuous reversibility is the differentiable unitary family delivered by phase-locking, exact in the idealized limit and holding at finite ϵ up to the residual dissipation $\mathcal{D}$, entering at second order in τ_S/τ_B ($\sim 10^{-64}$), which is a prediction of the framework rather than an approximation artifact; and tomographic locality rests on the derived composite factorization (§3.4, remark (ii)) together with the complex Hilbert space the correspondence delivers, so that derivation — including its open step, the unification with the staggered spin structure of the lattice substratum — is load-bearing for this hypothesis as well. Hypotheses of the reconstructions not on this list (perfect distinguishability, ideal compression, the subspace axiom) hold in the emergent theory only as properties of the output and are not claimed as independent corroboration. Given the independently sourced features, the reconstruction theorems force quantum theory uniquely; in particular the representation freedom of the correspondence (scope condition (i) above) cannot move the emergent description outside quantum theory — any operational theory with the marginalized process's features is quantum mechanics up to the standard gauge. The same division holds for the Born rule specifically: the constructive derivations above (equilibrium of the indivisible process; partial-trace structure) are joined by Gleason's theorem [J. Math. Mech. **6**, 885 (1957)], by which every noncontextual σ-additive probability measure on the projection lattice of the emergent Hilbert space, of dimension ≥ 3, has the Born form — the noncontextuality hypothesis being met structurally, because the emergent probabilities descend from a single measure over substratum microstates — with the envariance and decision-theoretic derivations [Zurek, Phys. Rev. Lett. **90**, 120404 (2003); Deutsch, Proc. R. Soc. A **455**, 3129 (1999); Wallace, *The Emergent Multiverse* (2012)] as convergent constructive members of the same family. The necessity statements add no new mechanism; their force is that the operational profile the construction produces has been independently proved to admit exactly one theory.

3.3 Bell Inequality Violations

The framework is a hidden variable theory — a deterministic completion of the kind the EPR argument concluded quantum mechanics required [46], with measurement revealing antecedently definite substratum facts — evading Bell's theorem [23] not by conspiratorial setting-dependence but by violating factorizability of the emergent joint process through P-indivisible joint dynamics — while remaining causally local. (The substratum-level statement, including in what precise sense this is not superdeterminism, is given explicitly below.) Two subsystems interacting at preparation carry a joint transition matrix $T_{QR} \neq T_Q \otimes T_R$. This non-factorizability *is* stochastic entanglement [11, 12, 24]. The idea that a shared background can induce bipartite entanglement has a physics lineage of its own: stochastic electrodynamics derived two-particle entanglement from coupling to the common zero-point field [47, 48]. The framework's shared hidden sector differs structurally — it is the derived complement of the observer's partition rather than a postulated field within spacetime, it sits prior to the emergent causal structure rather than inside it, and it carries the unistochasticity theorem and the derived Tsirelson bound rather than a model-by-model reconstruction — but the lineage is real and is acknowledged. Since the process is indivisible, no well-defined intermediate conditional probabilities permit Bell's factorization.

In the Jarrett decomposition of the emergent process, the framework violates outcome independence while preserving parameter independence and — at the operational level, where it is defined and testable — measurement independence: precisely the class Fine's theorem [25] permits. Barandes, Hasan, and Kagan [24] prove the maximum CHSH correlator is exactly Tsirelson's bound $2\sqrt{2}$, with independent support from Le et al. [26]. The bound also follows from the Stinespring route of §3.2, which establishes full unitary QM on a tensor-product Hilbert space; Tsirelson's original operator-norm argument [27] then yields $|\langle S \rangle| \leq 2\sqrt{2}$; achievability follows because all quantum states, including maximally entangled ones, are realizable in the emergent description.

The substratum level, stated explicitly. The Jarrett classification above is a statement about the emergent process, where the hidden variable is the process's state. It is natural — and a referee is right — to ask what becomes of Bell's premises when the hidden variable is taken instead to be the framework's deepest one: the substratum microstate $\lambda = (x, h)$ at preparation. At that level the framework is deterministic and causally local, and for a deterministic local theory factorizability *at fixed* λ is a theorem, not an independently deniable assumption; if measurement independence also held at that level, the CHSH bound of 2 would follow. It does not hold there — and cannot, for a reason that involves no conspiracy: in a closed deterministic universe the measurement settings are themselves functions of the total state, so the fixed-λ, varied-settings ensemble over which Bell's factorization integrates is not an ensemble the theory contains. The same move applies to Wigner's-friend-type paradoxes such as Frauchiger–Renner: the premise is not violated but *uninstantiable* — the framework's structure does not contain the object the assumption is about. What replaces it is the operational form of measurement independence — the empirically testable statement, about actual ensembles of runs, that pair-preparation degrees of freedom are uncorrelated with setting-generating degrees of freedom — and this holds dynamically rather than by stipulation: the substratum's measured chaotic mixing (two-trajectory Lyapunov measurement, $\lambda_{\text{max}} > 0$; spectral continuity, thermalization, cycle-structure consolidation, and orbit diagnostics concur, with the quantitative rate ensemble-dependent) actively destroys residual λ–setting correlations across macroscopic separations. The framework is therefore not superdeterministic in the sense that matters: superdeterminism *as an explanation* of Bell correlations is the position that λ–setting correlations do the explanatory work, and here they do none — they are dynamically suppressed, the correlations are produced by the indivisible joint dynamics, and the CHSH value is the stable structural output (exactly the Tsirelson bound) rather than a tuned coincidence. The no-fine-tuning analysis below makes the last point precise.

Fine-tuning and causal structure. The framework does not claim to restore Bell locality: outcome independence is genuinely violated. What it provides is a *derivation* of this nonlocality from the causal partition, rather than treating it as axiomatic. The fine-tuning objection (Wood and Spekkens, 2015) assumes DAG causal structure with Markov factorization; P-indivisible processes violate the Markov condition on any DAG, so the appropriate framework is the process tensor [7], within which no-signaling follows from marginalization structure without fine-tuning.

Parameter independence from spatial locality. Let V_A and V_B be spacelike separated visible-sector subsystems with disjoint coupling neighborhoods on G_φ. By range-1 coupling on the bounded-degree coupling graph (Lemma 1), the dynamics has the *spatial Markov property*: for any region V with interior V°, boundary ∂V, and exterior V^c, the next-step state at sites in V° depends only on current-step values within $V \cup \partial V$, so $V^\circ \perp V^c \mid \partial V$ in the joint distribution. Applied to V_A and V_B with disjoint coupling neighborhoods, this makes them conditionally independent given boundary data. Alice's measurement setting a determines which function she evaluates on V_A; it does not alter the transition probabilities for V_B, which depend only on V_B's local coupling neighborhood. Hence $P(x_B \mid a, b) = P(x_B \mid b)$ — parameter independence holds structurally for all bounded-degree graph topologies with range-1 coupling, with no fine-tuning of the hidden-variable distribution. Combined with the outcome-independence violation from stochastic entanglement, this gives the precise Jarrett decomposition.

3.4 The Characterization Theorem: Necessity of the Conditions

The logical structure: Barandes' correspondence gives QM $\iff$ P-indivisibility. What remains is P-indivisibility $\implies$ embedded observation under (C1)–(C3).

Dilation existence. Any single one-step doubly stochastic matrix on a finite configuration space can be realized as marginalization of a deterministic process on a larger state space — furnishing Lemmas 1–3. The scope is the one scoped at the lemma above: the matrix, not prescribed multi-time trajectory statistics.

Theorem (C1 equivalence). *The marginalized process is P-indivisible iff T*

is not a permutation matrix.

Proof. Forward: §2.3. Reverse: if T is a permutation, $\Lambda(k_2, k_1) = T^{(k_2-k_1)}$ is a valid stochastic matrix for all $k_2 > k_1 \Rightarrow$ P-divisible. □

Theorem (C2 necessity, conditional on ETH). *Assume the hidden-sector Hamiltonian H_H satisfies the eigenstate thermalization hypothesis (ETH) [17, 18]. Then in the fast-bath regime ($\tau_B \ll \tau_S$), the hidden sector dephases to its diagonal ensemble between coupling events, so the marginal dynamics on $\mathcal{C}_V$ is Markovian on accessible timescales ($T^{(k)} = T^k$) and hence P-divisible. Contrapositively, observable P-indivisibility in an ETH hidden sector requires $\tau_S \ll \tau_B$.*

Proof. Between coupling events (separated by τ_S), the hidden sector evolves under its own Hamiltonian H_H. The substratum dynamics is reversible, so there is no relaxation in the strict sense — finite-dimensional Hamiltonian flow has purely imaginary spectrum and Poincaré recurrence at t_R exponentially large in the system size. What does occur, under the ETH assumption, is *dephasing*: starting from any distribution $\mu_H(\cdot|x_i)$ conditioned on the previous visible state, the off-diagonal matrix elements in the energy basis decay on the timescale $\tau_B \sim \hbar/\Delta E$, where ΔE is the hidden sector's typical energy spread. After dephasing, the diagonal ensemble agrees with the microcanonical distribution to corrections exponentially small in the hidden-sector size, so for any few-body visible-sector observable O_V and any time $\tau_S \gg \tau_B$:

$$\left|\langle O_V\rangle_{\mu_H(\tau_S|x_i)} - \langle O_V\rangle_{\mathrm{eq}}\right| \;\lesssim\; \tau_B/\tau_S$$

In the fast-bath regime ($\tau_B \ll \tau_S$) this is parametrically small: from the visible sector's perspective the hidden sector "looks" thermalized at every coupling event regardless of its post-coupling state, even though the underlying trajectory remains reversible and quasi-periodic with recurrence only at $t \sim t_R$. Each single-step transition matrix T is therefore computed against the same effective equilibrium distribution, so $T^{(k)} = T^k$ — a homogeneous Markov chain on accessible timescales $t \ll t_R$, hence P-divisible in the observable sense.

Contrapositively: observable P-indivisibility requires $\tau_S \lesssim \tau_B$ — the visible sector must couple back into the hidden sector before dephasing erases the memory. For the strong, persistent P-indivisibility of the characterization theorem, the separation must be $\tau_S \ll \tau_B$. The continuous-time extension to general Hamiltonian flow replaces the collision-model picture with the full ETH/dephasing condition; the relevant machinery for chaotic open quantum systems in the weak-memory regime is established in [3, 8, 17, 18]. □

Remark (Status of ETH). *The eigenstate thermalization hypothesis is a well-supported conjecture, not a theorem, for generic chaotic many-body Hamiltonians. It is proved for specific models (random matrix ensembles, certain integrable-chaotic transitions) and supported by extensive numerical evidence across many systems [17, 18]. For the OI framework's hidden sector — the* *cosmological-horizon complement, a generic chaotic many-body system — ETH is the standard assumption. The C2 necessity direction is therefore conditional on ETH; relaxations or failures of ETH would require a separate argument. For the framework's specific substratum, moreover, the chaoticity underlying ETH is not merely the standard assumption: the realized bijection's discretization nonlinearity is numerically demonstrated to be chaotic — two-trajectory Lyapunov measurement, $\lambda_{\max} > 0$; spectral-continuity, thermalization, cycle-consolidation, and orbit diagnostics concur, the quantitative rate being ensemble-dependent — so the hidden sector is a measured-chaotic system rather than an idealized one, with the analytic characterization of its cycle structure the open residual recorded there. The C1 and C3 necessity theorems do not depend on ETH.*

Remark (Consequence for the characterization theorem). *The full biconditional (i) ⇔ (ii) ⇔ (iii) of the characterization theorem below is similarly conditional on ETH for the (ii) ⇒ (iii) direction specifically. The other directions — (i) ⇔ (ii) from [11, 12]; (iii) ⇒ (ii) from §2.3; C1 and C3 necessity from their own proofs — are unconditional.*

Theorem (C3 necessity). *Let $m = |\mathcal{C}_H|$. The non-Markovian mutual information satisfies:*

$$I(X_{<t}; X_{>t} \mid X_t) \le \log_2 m$$

Proof. The total system is deterministic: $X_{>t}$ is a function of (X_t, H_t). Conditioning on X_t: $X_{<t} \to H_t \to X_{>t}$ is a Markov chain. By data processing:

$$I(X_{<t}; X_{>t} \mid X_t) \le I(X_{<t}; H_t \mid X_t) \le H(H_t \mid X_t) \le \log_2 m \quad \square$$

Corollary (per-process capacity bound). Any deterministic-bijection realization of a process whose conditional past–future information reaches $I^* = \sup_t I(X_{<t}; X_{>t} \mid X_t)$ requires $m \ge 2^{I^*}$: the hidden sector must be at least as large as the backflow it carries. The bound is process-quantitative and does not reduce to the visible count — P-indivisibility alone does not force $m \ge n$: a two-hidden-state, three-visible-state bijection already exhibits full distinguishability revival ($m = 2 < n = 3$), so the necessity content of (C3) is the capacity to hold the *observed* backflow, not a bound set by n. Sustained backflow at rate $I_0 > 0$ per coupling event over K events requires $m \gtrsim 2^{KI_0}$ *under the additional hypothesis* that the per-event contributions accumulate without hidden-state reuse or compression; that independence hypothesis is not a consequence of P-indivisibility, and without it only the per-process bound above is forced. The framework's qualitative (C3) ("hidden sector large enough") is the conservative restatement of the per-process bound. □

The revival counterexample and the doubly-stochastic obstruction of §3.2 are certified by the deterministic probes in `papers/oi_lattice_code/foundations/` (`c3_dilation_probes.py`).

Definition (unitarily evolving QM). A stochastic process $\mathcal{S}$ on a finite configuration space $\mathcal{C}_V = \{x_1, \dots, x_n\}$ is *mathematically equivalent to unitarily evolving quantum mechanics* if there exists a Hilbert space $\mathcal{H} = \mathcal{H}_V \otimes \mathcal{H}_{\text{anc}}$ of total dimension $\leq n^3$, a Hermitian operator $\hat{H}$ on $\mathcal{H}$, a unitary family $U(t) = e^{-i\hat{H}t}$, and orthogonal projectors $P_i = |i\rangle\langle i| \otimes \mathbb{1}_{\text{anc}}$ onto the configuration sectors, such that the transition probabilities are the ancilla-uniform marginal of the dilated Born-rule probabilities: $T_{ij}(t) = \frac{1}{m_a} \sum_{k,l} |\langle j, l| U(t) |i, k\rangle|^2$ with $m_a = \dim \mathcal{H}_{\text{anc}}$. (This is the form the correspondence delivers — the visible matrix is the ancilla-marginal of the dilated unistochastic matrix, per the scope statement in §3.1. The strict form $T_{ij}(t) = |U_{ij}(t)|^2$ on visible indices is the trivial-ancilla special case $m_a = 1$ and only that case: a doubly stochastic marginal need not be unistochastic at every t, let alone realized by a single one-parameter unitary group on $\mathcal{H}_V$.) This definition captures the Hilbert space, unitary dynamics, and the Born rule. Additional quantum-mechanical structures — the tensor product decomposition for spatially separated visible-sector subsystems, state update, and the measurement formalism — are addressed in the remarks following the theorem.

Characterization theorem. *Let $\mathcal{S}$ be a stochastic process on a finite configuration space $\mathcal{C}_V$ with $|\mathcal{C}_V| \geq 2$, and consider dynamics on accessible timescales $t \ll t_R$ where t_R is the Poincaré recurrence time of any realizing substrate. Assume the hidden sector of any candidate realization satisfies ETH [17, 18]. Then the following are equivalent:*

(i) $\mathcal{S}$ is mathematically equivalent to unitarily evolving QM (in the sense of the preceding definition).

(ii) $\mathcal{S}$ is P-indivisible on accessible timescales — i.e., information backflow occurs at $t \ll t_R$, not merely at the formal Poincaré recurrence time.

(iii) $\mathcal{S}$ arises from marginalizing a deterministic bijection on $\mathcal{C}_V \times \mathcal{C}_H$ with (C1) non-trivial coupling, (C2) slow-bath memory ($\tau_S \ll \tau_B$), and (C3) sufficient capacity.

Proof. (i) $\iff$ (ii): Barandes [11, 12], on accessible timescales by construction (the equivalence is formulated for the dynamics observers can resolve, not for behavior at recurrence times). (iii) $\implies$ (ii): §2.3, where the accessible-timescale lemma upgrades the bare recurrence-based P-indivisibility (which follows from C1 alone) to information backflow on observable timescales using C2 and C3. (ii) $\implies$ (iii): the necessity theorems above — C1 from the equivalence with non-permutation T, C2 from the dephasing argument (without C2, T becomes Markovian on accessible timescales and information backflow disappears), C3 from the data-processing bound in its per-process form ($m \geq 2^{I^*}$, matching the backflow the process exhibits). □

QM is not merely *compatible with* embedded observation — it is *equivalent to* it.

Remark (scope of the equivalence). The characterization theorem establishes the Hilbert space, unitary dynamics, and Born-rule transition probabilities. Four further structures of operational quantum mechanics merit comment.

(i) Tensor product structure (visible–hidden). The Stinespring route (§3.2) constructs $\mathcal{H} = \mathcal{H}_V \otimes \mathcal{H}_H$ and derives the CPTP channel Φ by partial trace. The visible–hidden tensor product is therefore given by the construction, not postulated.

(ii) Tensor product structure (visible-sector subsystems). Spatially separated subsystems within the visible sector — e.g., two laboratories — correspond to subsets $V_A, V_B \subset V$ with disjoint coupling neighborhoods on the coupling graph. The spatial Markov property of range-1 dynamics (§3.3 above) guarantees conditional independence: $\rho_{AB} = \rho_A \otimes \rho_B$ conditioned on boundary data. The emergent Hilbert space inherits the factorization $\mathcal{H}_{V_A} \otimes \mathcal{H}_{V_B}$ from the locality of the coupling graph, with entanglement between A and B arising from shared boundary history — precisely the stochastic entanglement of [11, 24].

(iii) State update, measurement, and multi-time predictions. Projective measurement in the emergent description corresponds to conditioning on a visible-sector outcome and re-marginalizing over the hidden sector. The Lüders rule $\rho \to P_k \rho P_k / \mathrm{Tr}(P_k \rho)$ is the quantum transcription of Bayesian updating on the classical substratum (Lemma 3). No independent measurement postulate is required. Multi-time correlation functions, weak values, and Leggett-Garg inequalities are standard calculations within the delivered formalism ($\mathcal{H}_V$, $U(t)$, Born rule); they require no additional postulates beyond what the characterization theorem provides. The multi-time statement carries one qualification. The correspondence is engineered to reproduce rooted two-time transition data, which under-determines multi-time joint statistics [22]; the substratum, by contrast, fixes a unique multi-time law. Agreement of the substratum's conditioned multi-time statistics with the quantum formalism's intervention-rooted predictions is supported by the same chaotic-mixing suppression argument as operational measurement independence (§3.3), and residual pre-division memory, if present, would appear as deviations from quantum multi-time predictions — a falsifiable texture of the same standing as the residual dissipation $\mathcal{D}$.

(iv) Classical non-Markovianity and quantum mechanics. The equivalence is not a claim that all P-indivisible processes are "secretly quantum" in some metaphysical sense; it is a theorem that the mathematical structures are identical. Classical systems exhibiting P-indivisibility — renewal processes, semi-Markov processes on finite state spaces — admit a unique unitary description (phase-locking lemma) that makes predictions beyond the transition data (energy eigenvalues, interference, entanglement), and can be realized as marginalization of deterministic dynamics under C1–C3 (necessity theorems). The framework does not distinguish "classical non-Markovian" from "quantum" — it proves they are the same category. The physical content of the unitary is not that of a mere mathematical embedding: it is the *specific* Hamiltonian uniquely determined by the transition data, arising from the *actual* dilation structure of the system (§3.2).

(v) The single load-bearing external dependency. The necessity direction imports one external result rather than proving it internally: the Barandes stochastic-quantum correspondence [11, 12], which establishes that a P-indivisible stochastic process on a finite configuration space and a unitarily-evolving quantum system on a finite Hilbert space are the same mathematical object (the "indivisible" and "unitary" descriptions are inter-translatable). Everything in the necessity direction downstream of this — the dilation construction, the C1/C3 necessity theorems, the ETH-conditional C2 necessity — is the framework's own, but the bridge identifying P-indivisibility with quantum structure is Barandes'. This means the necessity direction has a clean falsification target distinct from the rest of the framework: if the Barandes correspondence were shown to be incomplete or to require additional hypotheses beyond P-indivisibility (e.g. a positivity or normalizability condition that not all P-indivisible processes satisfy), the necessity direction would inherit those hypotheses and weaken accordingly. The framework takes the correspondence as established in [11, 12] and does not reproduce its proof; a referee skeptical of the necessity direction should direct that skepticism first at the Barandes correspondence, since that is the load-bearing import, and the framework's own contributions are conditional on it. The sufficiency direction (C1–C3 $\Rightarrow$ QM) uses the correspondence in the same way but is independently supported by the explicit dilation construction of §3.2, so a weakening of Barandes would affect necessity more than sufficiency.

3.5 Generative and limitative faces of the observation map

The characterization theorem establishes what embedded observation *produces*. The same trace-out map has a complementary structural property — what it *fails to fix* — and stating it makes precise a limitation that recurs in the gravitational and Standard-Model sectors (the residual-input status of spatial isotropy in the framework's gravitational and Standard-Model treatments).

Lemma (rest-frame selection). *Let the embedded-observation partition $S = V \sqcup B$ be realized across a causal horizon $\mathcal{H}$, with induced coarse-graining map T. The horizon fixes a simultaneity foliation of the substratum, hence a distinguished timelike direction n (the frame in which the update steps are simultaneous). Then the invariance group of T within the local frame transformations $O(3,1)$ is the stabilizer of n — the spatial rotation group $O(3)$, the little group of a timelike vector. Consequently local frame transformations split into those preserved by T (spatial rotations, which fix n) and those not preserved by T (boosts, which move n).*

Proof. T is defined by the horizon partition, which fixes the simultaneity foliation and hence n up to sign. A frame transformation $\Lambda \in O(3,1)$ is an invariance of T only if it preserves the data defining T, i.e. only if $\Lambda n = n$. The stabilizer of a timelike vector in $O(3,1)$ is isomorphic to $O(3)$ (the standard little-group classification). Boosts map n to a distinct timelike vector and so are not invariances;

spatial rotations fix n and so are. $\square$

Corollary (boost/rotation inequivalence). Because T selects the timelike direction n but no spatial direction, the emergent theory inherits from the map a physical rest frame but no physical spatial axis. Lorentz-violating operators therefore have structurally inequivalent status by sector: boost-sector violation is tied to the map-selected object n (and is correspondingly exposed to map-level structure — the local-boost-invariance prerequisite of the framework's horizon-thermodynamic construction, the custodial mechanism of its lattice Standard-Model construction), whereas rotational-sector violation is referred to no map-selected object and is inherited from the substratum constitution alone, where the cubic point group protects it only at tree level. This is the structural origin of the boost/rotation asymmetry recorded in the framework's Lorentz-violation accounting; it explains the asymmetry without resolving either sub-sector.

Converse, and the resulting biconditional. The lemma admits a converse, which upgrades it to an equivalence and sharpens the corollary. *If the invariance group of T within $O(3,1)$ is a maximal compact subgroup (isomorphic to $O(3)$), then it is the stabilizer of a unique timelike direction n, so T selects a rest frame.* This is the standard maximal-compact-subgroup structure of the Lorentz group: the maximal compact subgroups of $O(3,1)$ are the conjugates of the rotation group $O(3)$, and they stand in bijection with the timelike directions (equivalently, with the points of the rest-frame space $O^+(3,1)/O(3) = H^3$), each subgroup fixing exactly the one-dimensional timelike line it leaves invariant. Combining the two directions:

> *The coarse-graining map T selects a unique preferred timelike direction n* ***if and only if*** *its invariance group within the local frame group $O(3,1)$ is a maximal compact subgroup $O(3)$.*

The content of the converse is not decorative. It says that **rotational invariance of the observation map and rest-frame selection are the same fact**: one cannot have T be rotation-invariant without its selecting a preferred frame that boosts violate, and conversely. The boost/rotation asymmetry is therefore not a contingent feature of how T happens to act — it is necessary. The residual rotational symmetry *is* the boost-breaking frame selection; spatial isotropy of the emergent description and the existence of a cosmic rest frame are two descriptions of one structure, not independent facts. (As with the forward direction, the converse rests on the maximal-compact-subgroup classification — standard Lie theory, not a framework-specific input — and requires the invariance group to be the full $O(3)$; it completes the equivalence at the same elementary depth as the lemma, not at the depth of the characterization theorem.)

The two faces, and a caution against over-reading them. The characterization theorem and this lemma are two properties of the *same* map T: what it produces (its image: the quantum kinematics) and what it leaves invariant (its symmetry: spatial rotations only). It is tempting to cast these as opposite operations — generation versus limitation — but they are the *same* operation, a projection of the observed statistics under information missing to the embedded observer; what differs is the *type* of missing information. In the quantum case the missing information is the hidden-sector *state*, which is dynamically resampled (the bath thermalizes) and therefore fluctuates; averaging over a fluctuating quantity is productive and yields the non-Markovian statistics that are quantum mechanics. In the isotropy case the “missing” information is the *orientation* of the lattice constitution, a fixed global parameter that is not resampled; there is no ensemble to average over, and the observer’s ignorance of the lattice axes is not a statistical mixture. The discriminant is whether the missing information fluctuates: fluctuating missingness is averaged into emergent structure, fixed missingness is inherited. Both faces trace to the temporal asymmetry of embedded observation — the slow-bath condition (C2) that drives the quantum emergence is the same feature that selects the simultaneity surface and hence the timelike n — but the quantum/isotropy contrast is properly attributed to fluctuating-versus-fixed missingness, not to a dichotomy of operations.

Two honest qualifications bound this subsection. First, the lemma is one-directional and near-elementary (it follows from the little-group classification once the horizon is granted to select n); it is far less deep than the characterization theorem, which rests on the Barandes correspondence, and it is labelled a lemma for that reason. Second, it is a structural statement about the map’s reach, not a derivation of a physical law: it explains *why* the framework can constrain the boost sector and must take spatial isotropy as input, but it neither supplies that isotropy nor forbids its violation. (Whether the fixed orientation could, in some régime, be effectively resampled — restoring isotropy generatively with a calculable residual anisotropy of the $\ell = 4$ cubic-harmonic form — is a separate open question pursued in the book’s open-problems treatment; the present lemma is agnostic on it.)

4.1 Interpretive Consequences

The degrees of freedom involved in quantum experiments — photons, electrons, slits, detectors — are all visible-sector objects. Their quantum behavior is a downstream consequence of the trace-out: the emergent quantum mechanics, once established by the theorem, governs all visible-sector dynamics.

In the double-slit experiment, the particle traverses a single slit in the deterministic substratum. The interference pattern arises because opening or closing the second slit changes the boundary conditions of the transition matrix, altering the distribution of detection events. A which-path detector at one slit couples the trajectory to additional visible-sector degrees of freedom, changing the transition matrix and eliminating the interference terms. The Everettian measure problem dissolves: the system evolves as a single reality; “branches” are features of the compressed description. The Born rule, often treated as an independent postulate, is derived: it is the equilibrium distribution of the indivisible stochastic process (§3.1), not an additional assumption.

Observer covariance and nested observers. The framework treats the observer partition $V \subsetneq S$ as a choice, not an absolute. Different partitions correspond to different observers; each partition satisfying the characterization conditions produces its own C1–C3-quantum description of the remaining degrees of freedom. This is not a multiplicity of incompatible physics — it is a covariance property of the framework under observer-partition choice, which resolves the standard nested-observer paradoxes.

Consider the Wigner's-friend setup: two disjoint partitions $V_W, V_F \subset S$ (Wigner and Friend) with $V_W \cup V_F \subsetneq S$, and shared external hidden sector $H = S \setminus (V_W \cup V_F)$. The framework assigns three descriptions, all derived from the same φ by the same trace-out procedure:

- **Friend's description.** Trace over $S \setminus V_F = V_W \cup H$. Under C1–C3 applied to $(V_F, S \setminus V_F)$, Friend's reduced description is quantum-mechanical. Friend observes definite outcomes in her visible sector, including any system she measures.
- **Wigner's description.** Trace over $S \setminus V_W = V_F \cup H$. Friend is part of Wigner's hidden sector. Under C1–C3 applied to $(V_W, S \setminus V_W)$, Wigner's reduced description is also quantum-mechanical. Wigner does not see a definite Friend-outcome within his own description — Friend's state is traced out.
- **Joint description.** The marginal transition matrix $T_{WF}(\alpha, \beta \to \alpha', \beta') = |\{h \in H : \pi_{V_W \cup V_F}(\varphi(\alpha, \beta, h)) = (\alpha', \beta')\}|/|H|$ exists and is unique given φ. Its marginals reproduce Wigner's T_W and Friend's T_F respectively.

The three descriptions are simultaneous rather than competing. Friend's claim that she has a definite outcome, and Wigner's description of Friend as a traced-out subsystem, are statements in different reduced descriptions, not conflicting claims about the substratum state. At the substratum level, Friend, Wigner, and the system all have definite configurations; neither Friend nor Wigner has epistemic access to the complete state, and this epistemic limitation produces their respective quantum descriptions through the same mechanism in §2–§3. The apparent "paradox" dissolves into a covariance property: observer identity is partition-gauge, not absolute, and framework predictions are covariant under choice of observer partition. No-go theorems assuming a single universal quantum description (such as Frauchiger–Renner) do not apply, because the framework provides multiple simultaneous reduced descriptions, each internally consistent with respect to its own partition, with all derived from the same deterministic substratum.

4.2 Scope

The characterization theorem is a full equivalence: QM $\iff$ embedded observation under (C1)–(C3). The theorem does not identify which physical systems satisfy the conditions; this is an empirical question. Universality in our universe follows from the shared causal partition defined by null geodesics.

4.3 The Status of the Discreteness Scale

ϵ does not smuggle in a quantum assumption, but the reason is foundational rather than field-theoretic. Finiteness of the *visible* configuration space is a lemma of the observation axiom itself: a single tokened differentiation registers determinate — and therefore finitely structured — content, so what is distinguishable at the locus per moment is finite; invoking holographic bounds at the foundational layer would be circular, those being downstream physical results. Finiteness of the hidden sector holds in the forward construction with a two-part status (A1, as scoped in the companion paper: the boundary layer $\mathcal{C}_B$ is bounded by the same finite-entropy argument that bounds $\mathcal{C}_V$, while the deep-sector cardinality beyond it is a gauge choice — the minimal finite representative under deep-sector enlargement), and is *constructed* rather than assumed in the reconstruction direction (§3.2's dilation supplies a finite hidden space). Holographic entropy bounds [2] then enter not as motivation but as a self-consistency check: the framework derives $\hbar$ and the Bekenstein–Hawking coefficient downstream, and the derived bound is compatible with the finiteness assumed here — a closed loop, not a circular one.

4.4 Relation to Prior Work

't Hooft [1]: Differs in mechanism (epistemic trace-out vs. information loss), generality (any embedded observer vs. particular rules), and Bell placement (outcome independence violation vs. superdeterminism). **QBism/relational QM:** Share observer-relative epistemic states; this framework provides a structural *why* and quantitative predictions.

Relation to prior stochastic-quantum programs. Three earlier programs sought to derive quantum mechanics from classical-looking substrates. **Nelson's stochastic mechanics [32]** postulates a continuous Brownian-motion-like process on configuration space with drift determined by the imaginary part of the log wave function; the Schrödinger equation follows from the Fokker-Planck structure. Wallstrom [33] showed that Nelson's program cannot uniquely select the quantization condition $\oint p \cdot dq = nh$ — the stochastic process admits a broader class of solutions than standard QM, and the quantization must be reimposed by hand. The present framework is not subject to this critique because it does not derive QM from a continuous stochastic process on ψ or on configuration space; it derives QM from the characterization theorem for embedded observers on a deterministic bijection, with stochasticity emerging from the trace-out over the hidden sector rather than being fundamental. **Adler's**

trace dynamics [**34**] derives QM from statistical mechanics of matrix-valued dynamical variables with a trace-class Hamiltonian, with the canonical commutation relations emerging as expectation values over a specific equilibrium ensemble. The framework here requires neither matrix-valued variables nor an equilibrium-ensemble derivation; the substratum is a finite configuration space with a deterministic bijection, and the emergence mechanism is the Stinespring dilation applied to observer-marginalized transition probabilities. **Barandes's stochastic-quantum correspondence** [**11, 12**], used as a technical tool in §3.1, is not a competitor but a building block: the correspondence maps P-indivisible stochastic processes on finite configuration spaces to unitary quantum evolutions on finite Hilbert spaces, and the present framework supplies the physical mechanism (embedded observation under C1–C3) that produces the P-indivisible processes Barandes then maps to quantum mechanics. The three earlier programs identified emergent QM as a legitimate research target; what the present framework adds is a full equivalence — necessary and sufficient conditions — and a constructive derivation chain that does not require independent quantization conditions or equilibrium ensembles.

Relation to the post-2022 observer-essentiality wave. A substantial body of work since 2022 has independently established that incorporating observers as physical degrees of freedom produces structural changes in the formalism of quantum field theory in gravitational settings, with results that converge on the foundational claim of the present framework — that observation cannot be treated as an external operation on physics — while differing in mechanism. **Chandrasekaran, Longo, Penington, and Witten** [**35**] showed that adding an observer to QFT in a gravitational subregion promotes the von Neumann algebra of observables from type III to type II_∞, with a well-defined trace and entropy; Witten [36] extended the construction to de Sitter space. **Maldacena's recent dS observer-cancellation** [**37**] demonstrates that the unphysical phase of the de Sitter sphere partition function cancels exactly when the observer-with-clock degrees of freedom are properly incorporated, removing a long-standing obstruction to a clean dS thermodynamic interpretation. **Harlow, Usatyuk, and Zhao** [**38**] argue that the closed-universe Hilbert-space dimension is determined by the observer's degrees of freedom rather than being a free parameter; the **AAIL construction** [**39**] provides a complementary calculation in which the relevant dimension scales as $d_{Ob} \cdot e^{2S_0}$, refining HUZ's heuristic bound. **DEHK partition-relativity** [**40**] establishes that the algebra of observables is partition-dependent and that physical predictions are encoded in equivalence classes under partition refinement, paralleling the present framework's commitment to partition-defined observation. **Slagle and Preskill** [**41**] constructed an explicit emergent-QM model in which boundary quantum dynamics arise from a classical bulk lattice with stochastic dynamics, demonstrating the logical possibility of QM-from-classical at the cost of an exponentially long extra spatial dimension and explicit stochasticity.

The present framework converges with these programs on the foundational substance — observation is not external to physics; the algebra of observables depends on the observer; the partition structure carries physical content — while differing in mechanism and scope. CLPW achieves type II algebra by adding observer degrees of freedom to QFT; the present framework derives the characterization of embedded observation directly from the deterministic substratum, with the type II structure as a consequence rather than a postulate. Maldacena requires an explicit observer-with-clock; here the partition trace-out is the *defining* operation, so the cancellation is structurally forced. HUZ argues that dimension is observer-determined; the present framework provides a microscopic realization in which the Hilbert-space dimension is computed directly from the partition geometry, with AAIL's $d_{Ob} \cdot e^{2S_0}$ formula matching the form of the substratum count. Slagle-Preskill demonstrates the logical possibility of QM-from-classical with an extra spatial dimension and stochastic dynamics; the present framework achieves the same emergence with neither — pure determinism on a finite substratum, no extra dimensions. The convergence of multiple independent programs on the foundational observer-essentiality move strengthens the empirical case for all of them; the present framework's distinguishing feature is that it derives this content from a finite deterministic substratum and, in doing so, produces a quantitative empirical record that the comparison programs do not currently match. The convergence is not a competition for priority — the deposit record is clear — but a triangulation: when several independent investigations of the foundations of physics arrive at structurally similar conclusions through different routes, the conclusions deserve more weight than any one of them alone would warrant.

Witten's Type II algebra program achieves finite entropy without finite-dimensional Hilbert spaces; this is compatible, since the framework's predictions depend on finite entropy (the physical content of Lemma 1), not on dimensional finiteness per se.

4.5 Assumptions, Limitations, and Falsifiability

The theorem requires Lemma 1, the stochastic-quantum bridge (§3.1 and §3.2; established by two independent routes), and genericity conditions (non-degenerate spectrum, non-vanishing overlaps), which hold for all but a measure-zero set. The P-indivisibility proof uses finiteness via the recurrence argument ($\varphi^N = \mathrm{id}$). The accessible-timescale backflow lemma (§2.3) provides a parallel route to P-indivisibility, establishing observable information backflow on the timescales accessible to laboratory experiments — physically indistinguishable from a timeless identity for all experiments well below the recurrence time.

Falsifiability. The framework makes a specific, experimentally testable structural claim: *the hidden sector is a classical substratum* — a finite set with a deterministic bijection. Every fundamental non-Markovian open-system process in nature — dynamics arising from embedded observation, not from engineered quantum environments — must therefore be classical-memory simulable in the operational sense of Bäcker, Beyer, and Strunz [28]: the environment's memory can be captured by a classical variable, and the non-Markovian dynamics reconstructed by conditioning on that variable. A physically realized fundamental open-system process whose memory provably exceeds the classical bound — demonstrated via the entanglement-structure protocols of [28, 29] — would falsify the framework's foundational ontology. This is a current-technology test whose experimental tools are under active development; a recent unified review [30] surveys the rapidly evolving classical-vs-quantum memory distinction in open-system dynamics. The claim is sharpest in the *C2-marginal regime* ($\tau_S \sim \tau_B$ rather than $\tau_S \ll \tau_B$), where non-Markovianity is $\mathcal{O}(1)$ rather than suppressed. Candidate physical realizations — strongly-driven superconducting qubits with two-level-system baths, biological electron transport with vibrational coupling, spin-boson dynamics at strong coupling — are the systems where the classical-memory simulability commitment is most stringently tested, and where emerging experimental and theoretical tools [28, 29, 30] are converging on the classical-vs-quantum memory distinction.

Theorem-level falsifiability. The characterization theorem itself would be falsified by a failure of both the stochastic-quantum correspondence and the Stinespring construction for the class of processes generated here — a possibility excluded by the independent proofs in §3.1 and §3.2. The classical-memory test above is the primary empirical falsifiability route, because the theorem is mathematical while the structural ontology commitment is physical.

Laboratory tests of the characterization theorem may be possible in tabletop systems where the hidden sector capacity is tunable.

The posit ledger. The two-axiom presentation of §1.2 names the foundation's primitives; the full argument consumes more inputs than two, and the complete list is collected here so it is countable in one place. The framework's substantive posits, beyond Axiom 1 (tokened differentiation, indubitable), are: (i) Axiom 2 — recurrence — owned as a posit with a non-derivability countermodel; (ii) finiteness of the *total* state space S — Lemma 1 delivers finiteness of the observer's distinguishable states, and the extension to all of S is A1 of the companion substratum construction — there scoped as the gauge choice of the minimal finite representative, the deep-sector cardinality being unobservable (deep-sector enlargement, [Substratum §4]) — consumed here by the recurrence step of §2.3 and by §4.6, whose conclusions accordingly hold on the finite representative and, being boundary-only in content, on every member of the gauge class (the transfer stated and proved as the corollary to Theorem 24 of [Substratum §4]); (iii) the record-preservation and eventual-image arguments that ground determinism (Lemma 3) close the hidden-sector margin using one *empirical* input — the observed unitarity of quantum dynamics — and one structural principle, so bijectivity is part-empirical rather than a pure lemma (a reader who declines the recovery may instead adopt reversibility as the standard information-conservation postulate of unitary quantum mechanics and classical Hamiltonian flow [42, 43], at no cost to the downstream results — see the remark following Lemma 3); (iv) the measure selection is a maximal-entropy principle ("the maximal-entropy invariant measure compatible with the observer's records") — a standard but substantive epistemic postulate, not derived; (v) ETH in the hidden sector, conditioning the C2-necessity direction, for a system class (emergent descriptions of linear-mod-q automata) on which the standard ETH evidence base is silent — this is the same open dynamical question as the chaotic-mixing suppression invoked for operational measurement independence (§3.3) and the typicality selection in (iv), one conditional wearing three hats; (vi) spatial isotropy, entering as empirical input E4 of the reconstruction; and (vii) linearity of the substratum update (A5), equivalent to amplitude-scale gauge invariance and owned at that more primitive location. Seven items, of unequal weight and mostly independently motivated. Part I of the companion methodology paper gives the complementary view: an eight-rung analysis of the primitive that recovers most of these as lemmas of the two-axiom base and isolates recurrence (Axiom 2) as the one irreducible non-lemma. The two accounts describe one structure — the rung analysis counts what is *recovered*, this ledger counts what is *assumed* — and agree on the substance, including that the determinism recovery closes its hidden-sector margin with an empirical input and that finiteness is delivered only per-moment. The two-axiom statement remains correct about what is *primitive*; this ledger states what is *assumed*, and the framework's claims should be read as conditional on the full list.

4.6 Structural Exclusion of Equilibrium-Phase Observers

An immediate corollary of (C1)–(C3) is a structural selection rule on where in a finite bijective (S, φ) observers can exist. The rule replaces a postulate often imported from elsewhere — the "past hypothesis" of low initial entropy — with a theorem derived from the framework's own definitions.

Setup. Let V be a partition with hidden sector $B = S \setminus V$. Define the coupling-graph neighborhood $\Sigma_{\text{loc}}(V)$ as the subset of S reached from V by at most $\tau_S/\Delta t$ steps of φ, where Δt is the update step. Let $\tau_{\text{mix}}(\Sigma)$ denote the mixing time of φ restricted to any φ-invariant subsystem Σ, set by the spectral gap of the transfer operator on Σ.

Theorem (structural observer selection). For any partition V of bounded coupling-graph diameter,

$$\tau_B(V) \leq \tau_{\text{mix}}(\Sigma_{\text{loc}}(V)) + O(\varepsilon),$$

where ε absorbs typical-set and finite-size corrections. In particular, if $\Sigma_{\text{loc}}(V)$ is in its local equilibrium macrostate in the sense of canonical typicality [31], $\tau_B(V)$ is bounded by the local thermal decorrelation time.

Proof. Correlations between a V-localized observable f and a B-localized observable g at time t require information to propagate from V to B at time 0 and back to influence the V-correlated component of g at time t. By the bounded coupling degree of φ, this round trip stays within $\Sigma_{\text{loc}}(V)$ for $t \leq \tau_S$. The spectral gap of $\varphi|_{\Sigma_{\text{loc}}}$ — which on an equilibrium microstate of $\mu|_{\Sigma_{\text{loc}}}$ sets the local thermal decorrelation rate — bounds the decay of such correlations exponentially at rate $1/\tau_{\text{mix}}$. Hence $\tau_B(V) \leq \tau_{\text{mix}}(\Sigma_{\text{loc}}) + O(\varepsilon)$. □

Corollary (anti-Boltzmann-brain). A Boltzmann brain is by construction a local fluctuation within a neighborhood Σ_{loc} that is itself in local thermal equilibrium — that is what "fluctuation from equilibrium" means. Applying the theorem: $\tau_B(V_{\text{BB}})$ is bounded by the local thermal decorrelation time $\sim \hbar/(k_B T_{\text{local}})$, which is at most $\sim 10^{-12}$ s for any temperature warm enough to support structured matter. Any finite-information observer has τ_S bounded below by its own internal processing time — for biological or synthetic BBs of any currently-envisioned type, $\tau_S \gtrsim 10^{-6}$ s. Condition (C2), $\tau_S \ll \tau_B$, therefore fails by at least six orders of magnitude for any BB. Since QM emergence under the framework requires (C2) as a necessary condition (§3.4), Boltzmann-brain observers do not see emergent QM; they are not observers in the sense the framework's definitions require. The exclusion is structural, not measure-theoretic — it follows from the incompatibility of (C2) with local equilibrium, not from arguments about the ratio of BBs to "ordinary" observers in an anthropic sample.

Corollary (past-hypothesis replacement). For (S, φ) in its typical orbit phase — equivalently, in its equilibrium macrostate — canonical typicality [31] implies every local subsystem is also in its local equilibrium. By the theorem, $\tau_B(V)$ is bounded by the local τ_{mix} for every bounded-diameter partition V. No partition supports (C2). The framework's observers therefore exist only in the non-equilibrium phase of (S, φ). The low-entropy initial condition commonly imported as a "past hypothesis" to select observer-compatible configurations is replaced here by a structural selection rule: observers exist where (C1)–(C3) are satisfiable, and (C1)–(C3) are not satisfiable in the equilibrium phase.

Scope. The theorem uses only bounded coupling, finiteness of S, and the canonical-typicality result [31]; no new axioms are introduced. Three scope notes. (i) *Locality of V.* The theorem assumes V has bounded coupling-graph diameter. Adversarially non-local partitions — cherry-picking distant subsystems that happen to be correlated — are excluded by the bounded coupling degree of φ and the finite response time of any physical observer. (ii) *Ergodicity.* The argument assumes the spectral gap on typical orbits of φ, which follows from statistical isotropy and bounded-coupling regularity; pathological non-ergodic φ (many short orbits) is excluded by the structural assumptions already in force. (iii) *The cosmological realization.* The de Sitter horizon is thermal in the static-patch sense at Hawking temperature, but the globally expanding system is not in equilibrium with respect to any static-observer partition. The horizon's expansion sets $\tau_B \sim H^{-1}$ structurally, not thermally, which is the OI-native mechanism making human-timescale laboratory observation compatible with (C2).

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5. CONCLUSION

The general theorem. Under a single definition — observation is a proper subsystem of a deterministic whole — and conditions (C1)–(C3), the P-indivisibility of reduced dynamics is proved: any bijective dynamics on a finite system with non-trivial coupling necessarily produces P-indivisible stochastic dynamics. By the stochastic-quantum correspondence, the observer's description is necessarily quantum mechanics. Continuous-time transition data resolves all gauge freedom. The Schrödinger equation, Born rule, and Bell violations are structural consequences. The characterization theorem establishes that the conditions are necessary: QM and embedded observation under (C1)–(C3) are equivalent.

Code and Data Availability

The verification code for §3.2 is under `papers/oi_lattice_code/coherence/`: `coherence_enumeration.py` (the (C1) counterexample, the exhaustive small-system enumeration behind the counts quoted in the Remark on the failure of (C1), and the Lemma 1 boundary bounds) `coherence_theorem_tests.py` (the Weyl transfer rule and normal form of Lemma 2, the separability threshold exhaustively at $s = 2, 3$, and both directions of Corollary 1 — a negative partial transpose wherever $w < \kappa$, an explicit measure-and-prepare form wherever $w = \kappa$ — for $q = 2$ and $q = 3$), and `eb_pindivisibility_tests.py` (the strict one-way containment of complete dephasing in entanglement breaking, and an explicit reversible dilation whose one-step channel is entanglement-breaking while its visible process is P-indivisible). All are deterministic, depend only on NumPy and SciPy, and exit non-zero on failure; no result in this paper rests on sampling or on a random seed.

The repository is archived on Zenodo with concept DOI 10.5281/zenodo.19060318, which resolves to the latest version; per-release version DOIs are minted at release time, and a reader reproducing a specific claim should cite the version DOI of the release carrying it rather than the concept DOI. Paths above are relative to the repository root, which is also the root of the archived snapshot. The code is also available on GitHub at https://github.com/amaybaum/incompleteness and is released under the MIT License.

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